

UNIVERSITY OF BONN

MASTER THESIS

Physical vs Digital Surveillance

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ABSTRACT

This study examines how people in the United States judge digital and physical surveillance across three lenses: effectiveness, invasiveness and acceptability. The preregistered survey ran May 27–June 3, 2025, included 33 main questions plus one attention check, and drew 1,089 starts, of which 982 met inclusion criteria. We tested 12 confirmatory hypotheses (H1–H12) and 16 exploratory hypotheses (EH1–EH16). Overt methods are more acceptable than covert and physical is slightly more acceptable than digital, yet digital is often judged more effective. Effectiveness predicts acceptance more than invasiveness. Many see surveillance as common but do not expect it to affect them personally. Trust in institutions is lower than prior studies and strongly shapes acceptance, and political affiliation is a clear divider.

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1. INTRODUCTION

Surveillance today is not just about grainy CCTV or a wiretap in a movie. It sits in our pockets, runs on our phones and powers the tools police use to make sense of the vast amounts of data present in everyday life. In the past year, the debate around the topic of surveillance has sharpened. In the UK, the government pressed Apple to consider a law-enforcement backdoor[21], with Sweden signaling similar interest[17]. In Germany, proposals to let police use facial recognition for counterterrorism test the edges of the EU’s AI rules[30], and Bavaria’s interior minister has called for live facial recognition in train stations[20, 27]. Across the Atlantic, the U.S. and Canada have expanded aerial monitoring along their borders with drones, helicopters, and fixed towers, aiming for round-the-clock coverage[7, 24]. The technology keeps moving and so does the policy. But a simple question remains: how do people feel about all this? What sentiments, worries and attitudes should one consider in policy making?

Much of the public conversation is led by groups with strong stakes: child-protection advocates, police, security agencies, and civil-liberties organizations or cybersecurity activists. All these provide valuable insights and perspectives on the topic and they frame the risks and benefits differently. Still, the everyday experience is more mundane and more mixed. We unlock phones with our faces. We pass camera domes in supermarkets. We read headlines about “AI policing” and also about data breaches. This all boils down to the question: Do people see digital and physical surveillance as the same thing, or do they feel different about a street camera than about a phone scan? And what shapes those views: trust in government, experiences with feeling targeted by authorities, education, or where someone lives?

This thesis looks in particular at *perceived invasiveness*: how far a measure feels like it crosses a personal line. Alongside that, it examines *acceptability* (is this okay and should it be allowed?) and *effectiveness* (does it work?). These ideas sound abstract, but they show up in simple, everyday judgments: Would you be comfortable with body-worn cameras? With location tracking on a phone during a criminal investigation? With scanning the cloud? By comparing answers across demographic groups (age, education, political context, and more), we can see where opinions converge, where they split, and what might explain the difference.

The stakes are practical. Public support or the lack of it, shapes whether policies can be implemented, how courts view proportionality, and how companies design products by default and what risk of problems and backlash implementations may have. If people accept overt cameras but balk at covert data collection, that matters for where resources go and where more conversation is needed. If trust in institutions moves alongside acceptance, then efforts to build transparency and oversight are not window dressing, they are part of the policy itself. And if people regularly mix general attitudes (“I don’t like surveillance”) with concrete exceptions (“... but I support body cams”), then the details of how a method is communicated can make a significant difference.

To keep the legal and political backdrop coherent, this study focuses on respondents in the United States. The approach is straightforward: a cross-sectional survey, fielded to a broad adult sample, that asks about general attitudes, method-specific judgments, and a handful of everyday scenarios. It pairs those judgments with perceptions of effectiveness and the felt risk of abuse, and it situates them in context: how much they trust various institutions, and how they judge local issues like crime or corruption. Randomization was used to soften order effects, and preregistered analyses keep the confirmatory core separate from the exploratory parts of the study.

The goal here is not to choose a side in the privacy–security debate but to describe how people currently weigh the trade-offs: where they draw the line, when they make exceptions, and what moves them. In short, this is a study about feelings and judgments in a world where surveillance is both ordinary and contested. The findings aim to offer a clearer picture of how citizens think about digital and physical

surveillance today and, in doing so, to inform policy, practice, and technology that better balance safety with individual rights.

2. RELATED WORK

The comparison of physical and digital surveillance falls into the wider privacy–security debate, a longstanding issue with extensive research on trade-offs in different contexts. For over fifty years these questions have been debated and evaluated [29, 31]. A significant body of research has explored how individuals perceive the trade-offs between security and privacy and what influences public acceptance of surveillance. Studies indicate that acceptance often depends on contextual factors such as perceived crime rates, trust in authorities, demographics and the framing of security measures.

In 2018, Kostka et al. [23] investigated public attitudes toward social credit systems in China and highlighted correlations between approval and demographics, wealth, and education, identifying high approval where systems are seen as improving quality of life. Xu et al. [19] reported a significant drop in approval once respondents learned about repressive potential, underscoring the role of information and framing. Karpa et al. [18] compared support for digital governance across Russia, Germany, Turkey, the United States and Estonia, finding that reliance on government-controlled information sources was associated with greater endorsement of such tools.

Lund-Tønnesen [25] compared surveillance and privacy regulation in Germany, Norway and the UK, tracing differences to historical, cultural and legal traditions. Germany is described as having one of the strictest privacy regimes, grounded in constitutional doctrine and strong enforcement, alongside high public sensitivity to surveillance. Norway, while valuing privacy, shows a more favorable public attitude toward surveillance, partly due to high social and political trust. By contrast, the UK has historically prioritized government and business data interests, taking a pragmatic approach to balancing surveillance and privacy.

Smith et. al have studied public attitudes toward security-focused applications such as the Corona-Warn-App in Germany, highlighting how the complicated relationship of trust in the government and public communication influence the population’s willingness to adopt the app[15][14]. Likewise, Smith et al.’s research on client-side scanning attitudes, also highlights how trust in authorities and companies affects the public’s willingness to accept surveillance technologies. They also note that the majority of participants who supported CSS were also worried about potential abuse[12].

The digital age has intensified these discussions, particularly with the increasing prevalence of mass surveillance technologies and the widespread collection of personal data. Lyon (2018) explores the normalization of surveillance in everyday life, arguing that surveillance in the digital age has evolved into a pervasive mechanism where particularly social media platforms can be used for data gathering and "even initiate surveillance"[26]. Further more, the integration of AI into surveillance systems has raised significant privacy issues. Clearview AI, for instance, developed a facial recognition software in 2020 that allows users to upload an image and match it against a vast database of online photos, providing links to where the matches appear on the internet. According to the company, more than 600 law enforcement agencies started using the application within the first year. The company’s practices, including scraping images from social media platforms without consent, have sparked global debates about privacy and the need for regulation in facial recognition technology[16]. Additionally, governmental use of AI for surveillance has also been contentious. The U.S. State Department, for example, announced plans to use AI to monitor foreign students’ social media for perceived support of Hamas, aiming to revoke visas based on AI-assisted reviews. Critics argue that such measures could lead to errors, privacy violations, and infringements on free speech rights[8]. Additionally, AI surveillance has been increasingly deployed for predictive policing and social control. Brayne [5] conducted a five-year long study during which they followed the artificial intelligence-led police professionals within the Los Angeles Police Department (LAPD). They raise the question whether crime predicting algorithms are able to anticipate individuals falling into criminality correctly, or if they rather perpetuate individuals in becoming criminals once

they have been ‘marked’ by the system. She states how data-intensive and artificial intelligence based, predictive policing practices render it increasingly impossible for surveilled, and apprehended citizens, to contest police suspicions and highlights the risks of algorithmic bias due to historical biases in the crime data resulting in over-policing of marginalized communities.

A growing area of concern in digital surveillance is the role of encryption and governmental access to private communications. The debate over encryption backdoors pits law enforcement agencies against privacy advocates, with the former arguing that backdoors are necessary to combat crime and terrorism, while the latter warn that such measures weaken security for all users, adding that any proposals that alter the security dynamics online should be approached with caution [2].

The existing literature provides valuable insights into how digital and physical surveillance are perceived and accepted across different societies. This study aims to build on these findings by examining how societal factors such as education, political climate, and historical experiences influence individuals’ trade-offs between security and privacy. Additionally, the research will contribute empirical data on public attitudes toward digital and physical surveillance in multiple countries, expanding on prior studies that have primarily focused on either national-level perspectives or specific surveillance technologies.

By integrating these perspectives, this study seeks to deepen the understanding of how digital and physical surveillance measures are viewed in different cultural and legal contexts, providing a comprehensive analysis of privacy-security trade-offs in the modern era.

3. BACKGROUND

Spearman correlation measures whether two variables move together in a consistent direction when their values are ranked. It replaces raw numbers with ranks, then computes the correlation on those ranks. This makes it robust to skewed distributions and suitable for Likert-type responses. A positive Spearman coefficient indicates that higher ranks on one variable tend to come with higher ranks on the other. A negative coefficient indicates the opposite. Ties are handled with a standard correction so the estimate remains unbiased.

Paired t-test evaluates whether the mean difference between two measurements taken from the same participants is different from zero. The test works on the set of within-person differences, computes their mean, and divides by the estimated standard error to form a t statistic. The key assumption is that these paired differences are approximately normally distributed. When that assumption is doubtful, the **Wilcoxon signed-rank test** provides a distribution-free alternative. It ranks the absolute differences, assigns the original signs, and tests whether positive and negative signed ranks balance out around zero.

One-way ANOVA tests whether three or more groups have the same mean on a continuous outcome. It partitions total variability into between-group and within-group components and uses their ratio to form an F statistic. Assumptions are approximate normality of residuals and similar variances across groups. If those do not hold, the **Kruskal–Wallis test** is used instead. It replaces values by ranks and assesses whether the rank sums differ across groups more than expected by chance. For two independent groups, the **Mann–Whitney U test** plays the same role as the independent-samples t test but on ranks.

Chi-square goodness-of-fit compares observed counts across categories to expected counts under a simple reference model such as a uniform split. It sums squared deviations between observed and expected counts scaled by the expected counts to form the χ^2 statistic. Large values indicate poor fit. When assessing whether two categorical variables are associated, the **chi-square test of independence** uses the same χ^2 logic on a contingency table. If any expected cell count is very small, **Fisher’s exact test** is preferred because it relies on the exact hypergeometric distribution rather than a large-sample approximation.

Holm–Bonferroni correction controls the familywise error rate when testing several hypotheses together. P-values are ordered from smallest to largest and each is compared to a progressively relaxed threshold $\alpha/(m - k + 1)$, where m is the number of tests and k is the p-value’s rank. The first non-significant comparison stops the sequence. This procedure is uniformly more powerful than simple Bonferroni while maintaining strong control of Type I error. **Benjamini–Hochberg** controls the false discovery rate. It orders p-values and finds the largest k such that $p_{(k)} \leq k\alpha/m$. All p-values up to that rank are called significant. This keeps the expected proportion of false positives among significant findings at or below α , which is useful in exploratory settings.

Effect sizes and intervals accompany these tests to quantify practical importance and uncertainty. Cohen’s d summarizes standardized mean differences for t-tests. η^2 or ϵ^2 describe the share of variance explained in ANOVA or its rank-based analogue. Cramér’s V summarizes association strength for chi-square tests. Rank-based r provides a simple effect metric for Wilcoxon or Mann–Whitney. Confidence intervals are reported where feasible to show the range of estimates compatible with the data.

POLITICAL CONTEXT DURING DATA COLLECTION

The data was collected from May 27, 2025 at 16:41 to June 3, 2025 at 16:54. To provide political context of this time, this window fell during the early months of President Donald J. Trump’s second, non-consecutive term, which began with his inauguration as the 47th president on January 20, 2025[28, 34]. Generally speaking, the public life in the United States can be seen as marked by a clear split in political support within the nation. High political salience around the topics of border enforcement, public order

and the role of technology in government was present. To provide a picture of the satisfaction of the country in this time with their government, one should note that nationwide demonstrations had taken place. On February 5, 2025 coordinated protests were held in cities across all states in response to the administration's early agenda and the broader Project 2025 policy platform[4]. In April 2025 further nationwide "Hands Off" events drew large crowds, reflecting elevated protest activity in the first half of the year[33]. Independent monitoring also recorded a sharp rise in immigration-related demonstrations across the country during early this time period[6].

Institutional changes proceeded in parallel. The Federal Bureau of Investigation saw a leadership transition with Kashyap "Kash" Patel sworn in as director in February 2025 after Senate confirmation[10]. The Department of Justice announced updated enforcement priorities in May 2025, including changes to guidance on corporate and white-collar matters[11].

The period also reflected continuing polarization. Supporters emphasized a mandate for stricter immigration control and more assertive public safety policy. Critics voiced concerns about civil liberties, the potential chilling effect of surveillance technologies, and the safety of minorities and non-citizens in a tense rhetorical climate. Reports from protests and civil-society groups pointed to anxiety among immigrant communities and other minority groups about targeted enforcement and speech, which in turn can shape attitudes toward government surveillance and trust in institutions[4][6].

For further context, President Joseph R. Biden Jr.'s term had ended on January 20, 2025[32]. His administration's final months included ongoing debates over AI governance, data privacy, and law-enforcement technology at federal and state levels, as well as ongoing dissatisfaction with the government from certain parts of the population. The transition from the Biden administration to the second Trump administration therefore bracketed the survey period with a clear change in policy direction and political tone. For a study of public views on surveillance, this timing matters. Perceptions of effectiveness, acceptability, and invasiveness often move with institutional trust, perceived procedural safeguards such as warrants or judicial oversight, and the immediacy of headline issues.

4. METHODOLOGY

The overarching goal of this thesis is to design, conduct and analyze a study in order to help understand the societal factors that shape individuals' attitudes toward digital and physical surveillance and differences in their perception. The research focuses on a comparative analysis and examining the public perceptions of U.S. adults regarding surveillance techniques, their invasiveness, effectiveness and acceptability. It aims to understand how individuals perceive and evaluate different forms of surveillance, what factors influence their acceptance and whether there are discrepancies. The methodology for this research comprises a structured survey that examines attitudes toward surveillance, its perceived commonness, its acceptability, and the role of trust in authorities and institutions. Additionally, demographic factors and crime perception are explored as potential influences on surveillance attitudes.

As this research falls in the broader spectrum of the privacy versus security debate, whenever possible questions will be formulated with the intent to ensure comparability to previous work done in this research group. This includes using partially the same questions as done in [12].

4.1. SURVEY DEVELOPMENT

4.1.1. RESEARCH QUESTIONS AND HYPOTHESES

The following confirmatory hypotheses and research questions guide this study:

- **RQ1:** How do individuals perceive the differences between digital and physical surveillance?
 - H1:** Participants will perceive differences in how much information digital and physical surveillance can gather.
 - H2:** Participants will rate the invasiveness of digital and physical surveillance differently.
- **RQ2:** What factors influence the perceived invasiveness and acceptance of the different surveillance methods?
 - H3:** Acceptability differs between overt (informed) and covert (uninformed) surveillance.
 - H4:** The perceived invasiveness of a surveillance type is associated with its acceptability.
 - H5:** The perceived effectiveness of a surveillance method is associated with its acceptability.
- **RQ3:** How do trust and perceptions of authority affect acceptance of surveillance and concerns about its misuse?
 - H6-8:** Trust in government, law enforcement, and companies is positively associated with acceptance of surveillance measures.
 - H9-11:** Trust in government, law enforcement, and companies is negatively associated with concern about surveillance misuse.
 - H12:** Perceived levels of local crime and organized crime are associated with surveillance acceptance.

Hypotheses H6-H11 are based on previous research[12] and thus a directioned hypothesis has been chosen.

In addition to our preregistered confirmatory hypotheses, we plan to possibly explore the following questions and relationships in the dataset. These analyses are hypothesis-guided but will be interpreted as exploratory, and any results we report will be described as such.

- EH1:** Participants will perceive differences in how common digital and physical surveillance are.

EH2: Preferences for surveillance method (digital vs. physical) are associated with perceived invasiveness.

EH3: The perceived likelihood of being surveilled is associated with acceptance of surveillance practices.

EH4: Belief that surveillance has increased, decreased, or stayed the same is associated with the acceptance of surveillance.

EH5: Participants who both believe surveillance has increased and say they are worried about this will differ in their acceptance score compared to other combinations of these two variables.

EH6: Belief that surveillance has increased is associated with a different likelihood of saying the evolution of surveillance the past five years is worrisome than believing it has stayed the same or decreased.

EH7: Reporting that one feels deliberately disadvantaged by the government or police is associated with trust in authorities and concern about Surveillance Abuse.

EH8: Among those who perceive both crime and corruption as high, support for surveillance will differ depending on whether corruption or crime is perceived as more concerning.

EH9: Perceived levels of corruption are associated with trust in authorities.

EH10: Among participants who perceive high corruption and report low acceptance of surveillance, the option of independent monitoring is associated with differences in support for surveillance compared to those who reported high acceptance

EH11: Some participants will express general disagreement with a surveillance type (e.g., covert digital) while supporting most corresponding specific methods, and vice versa (general agreement but rejecting specific methods).

EH12: Some participants will express a preference for being surveilled through the method (digital or physical) that they also believe would reveal more information about them.

EH13: Age, gender, and education level are associated with differences in acceptance, trust, and perceived invasiveness of surveillance methods.

EH14: Participants with higher income or higher education will differ in their acceptance of digital surveillance methods.

EH15: Participants whose stated political affiliation aligns with the party currently in power (Republican) will differ in their trust in government, acceptance of surveillance, and concerns about surveillance abuse compared to participants whose affiliation does not align with the governing party.

EH16: Among participants who indicate their responses would change if the Democratic Party were currently in power, the type and direction of reported change (e.g., trust, support, fear of abuse) will differ based on their stated political affiliation.

4.1.2. SURVEY DESIGN

This observational study is a cross-sectional, between-subjects survey conducted using Qualtrics as the survey platform and Prolific for recruiting a representative sample of 1,000 U.S. adults.

4.1.2.1. QUESTIONNAIRE

It consists of 33 main questions and one attention check, structured into thematic blocks. The full questionnaire can be found in Appendix A. The general structure is as follows:

Welcome, Introduction and Consent Paragraph

General Attitudes Toward Surveillance: Participants evaluate whether U.S. law enforcement should be allowed to conduct covert/overt and digital/physical surveillance (Q1), and how effective each of these types is at preventing crime (Q2) using five-point likert scales.

Perceived Invasiveness, Awareness and Abuse Concerns: Respondents compare the invasiveness of digital vs. physical surveillance (Q3a/Q3b), rate how often they think about privacy in daily life (Q4),

and choose between digital or physical surveillance when forced to decide (Q5). They indicate which surveillance method they believe gathers more information (Q6), and how likely they believe it is for digital or physical surveillance to be abused by law enforcement (Q7–Q8).

Perceived Commonness and Self-Relevance: Questions assess perceived surveillance prevalence (Q9), personal likelihood of being surveilled (Q10), and which method would be more likely used on them (Q11).

Trends and Worries Over Time: Participants evaluate how surveillance has changed over the past five years (Q12), which type changed more (Q12.2/Q12.3), and whether they find these changes worrisome (Q13).

Environmental Factors and Perceived Threats: Respondents rate local levels of crime, corruption, and organized criminality (Q14), and indicate whether they believe surveillance serves to protect or control (Q15).

Trust in Institutions: Trust in government, agencies, companies, and fair treatment by law enforcement is assessed using agreement statements (Q16–Q19).

Surveillance Method Acceptability: Participants rate the acceptability of a broad range of digital and physical surveillance techniques (Q20), including AI tools, GPS tracking, malware, CCTV, drones, etc.

Follow-Up Explanations (Conditional): For contradictions or notable patterns in responses, participants are asked to explain their reasoning in free-text fields (Q21.1–Q21.3). Additionally, for the participants who perceived crime or corruption to be high, follow-up questions were asked on whether law enforcement lacks tools to address crime or organized crime (Q21.4–Q21.5), and which additional methods might be justified (Q21.6), support for third-party monitoring in cases of corruption (Q21.7) and trade-offs between corruption and crime concerns (Q21.8).

Minority and Discrimination Perceptions: Participants self-identify as minorities (Q22), and if applicable, report perceived discrimination by various institutions (Q22.1).

Political Context and Hypotheticals: Respondents indicate whether their answers would differ under a different political party's control (Q23), and in which ways their responses would shift (Q23.1–Q23.2).

Demographics: Standard demographic questions cover age (Q24), gender (Q25), location (Q26), nationality (Q27), ethnicity (Q28), work sector (Q29), tech affinity (Q30), political preference (Q31), education (Q32), and income (Q33).

4.1.2.2. RANDOMIZATION

To reduce potential order effects and formulation bias, randomization was implemented at multiple levels throughout the survey. First, participants were randomly assigned to one of two groups that differed in the order of key question blocks. One group was first presented with general acceptability ratings of covert and overt surveillance (Q1), followed by detailed descriptions of specific surveillance methods (Q20). The other group received the reverse order. This between-subjects randomization was designed to test for priming effects and ensure that general attitudes were not unduly influenced by preceding exposure to concrete examples.

Within-subject randomization was also applied where appropriate. For categorical questions with multiple sub-items, the order of items was randomized to mitigate response pattern biases. However, where a natural or ordinal structure existed order was kept fixed.

In addition, randomization was applied at the level of question wording to control for formulation bias. Q3a and Q3b presented comparative invasiveness questions in two different phrasings: one framing digital surveillance as the reference point and the other framing physical surveillance as the reference point. Each participant was shown only one of these formulations, randomly assigned using simple randomization.

All randomization procedures were implemented using Qualtrics' built-in randomization tools, which rely on simple random assignment without replacement.

4.2. SURVEY DEPLOYMENT

4.2.1. SAMPLING AND INCLUSION CRITERIA

The survey is conducted via Qualtrics, with participants recruited through Prolific using a representative sampling approach for U.S. adults based on age, gender, and ethnicity. Inclusion criteria include being 18 or older, giving consent to participate in the study, currently residing in the United States, being fluent in English and passing Prolific’s pre-screening for representative eligibility. Exclusion criteria include failing the attention check, being flagged as a bot, providing incomplete responses or completing the survey in less than 1/3 of the median completion time. As it is required of participants to answer all questions, we do not need to define how to handle missing data.

The survey takes approximately 10 minutes to complete. Each participant is compensated at a fair hourly rate, in line with or exceeding Prolific’s minimum pay guidelines. Recruitment and data collection are expected to be completed within less than a week, depending on participant flow and quality screening. All data will be reviewed after collection to exclude invalid responses before analysis.

4.2.2. POWER ANALYSIS

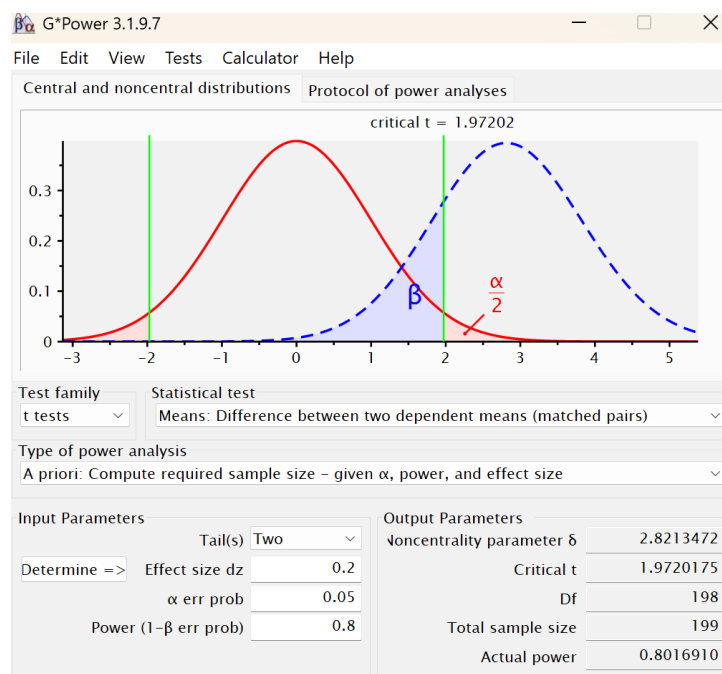


Figure 4.1.: Calculation of Sample Size for a two-tailed paired T-test using G*Power

For the various hypotheses in this study, a power analysis was conducted to determine the necessary sample size to ensure reliable results, considering effect size, significance level (0.05), and desired statistical power (0.8). The hypotheses are tested using different statistical methods based on the nature of the data and research question. The power analysis is done with G*Power[9]. Relevant tests for the power analysis are a paired t-test, a Spearman’s rank test, and a chi-square test with three categories.

Paired t-test: This test compares the means of two related groups to determine if there is a statistically significant difference between them. It is used for continuous data and the two groups are connected. For a medium effect size($d=0.5$) a sample size of 34 is required, for a small effect size ($d=0.2$) 199.

Spearman's rank test: This test is used for measuring the strength and direction of a monotonic relationship between two ordinal variables. For a medium effect ($r=0.3$) size, 84 participants are needed, and for a small effect size ($r=0.1$), about 782 participants.

Chi-Square Test: This test examines the relationship between categorical variables and is used to test whether the distribution of observed frequencies differs from the expected frequencies under the null hypothesis. It's ideal for analyzing how participants categorize information, in our case for three categories. For a medium effect size, 108 participants are required, and for a small effect size, around 964 participants.

Given the power analysis, the final target sample size is around 1000 participants, to account for unusable or incomplete responses. This sample size will provide sufficient statistical power (0.8) to detect meaningful differences or correlations across all hypotheses and tests, ensuring robust and reliable results for this study.

4.3. ADHERENCE TO SCIENTIFIC STANDARDS AND ETHICAL COMPLIANCE

This study adheres to common ethical guidelines to ensure both participant protection and data integrity. Clear information about the study's purpose, data usage and participants' rights is provided at the beginning of the study. All participants are required to provide informed consent before taking part in the survey, confirming that they agree to be part of the study and are over 18. They are informed about the purpose of the study, the use of collected data, and that they have the right to withdraw at any time for any reason. No personal information that can identify a person will be collected and responses will remain anonymous. To protect participant anonymity, the recruitment process will be conducted through a market research firm which does not have access to the survey data. The survey itself will be hosted on a secure platform, separate from the recruitment process, ensuring that identifying information is not linked to survey responses. Data that is automatically collected by the platform, such as IP addresses, is not stored and removed from the data. All demographic questions include an option for participants to decline to answer (e.g., "I prefer not to say"), allowing individuals to participate without feeling pressured to disclose personal information. They are informed that choosing not to answer certain questions will not result in any disadvantages for them. Furthermore, the study is reviewed by an institutional ethics board to confirm compliance with ethical standards.

To ensure the transparency and reproducibility of this study, the research hypotheses and the study outline are preregistered on the Open Science Framework (OSF)[1] before data collection begins. Preregistration helps prevent p-hacking, selective reporting, and other questionable research practices by committing to a predefined analytical approach. The statistical analysis will circumference the predefined research hypotheses and any changes or additional findings will be reported as such. This enhances the credibility of the study by ensuring that findings are interpreted within the originally intended framework rather than being shaped post hoc. A copy of the preregistration can be found in Appendix B.

During the survey development phase, established best practices for questionnaire design were followed where possible. For data collection and analysis, methodological standards will be adhered to. This will include transparent exclusion criteria for invalid responses (speeders, incomplete responses, failing attention checks or AI detection, see Section 4.2.1). Additionally, the conducted power analysis has determined an appropriate sample size, reducing the risk of underpowered or overfitted results. Applying predefined statistical tests as outlined in the preregistration will avoid data dredging and reporting effect sizes and confidence intervals alongside p-values will provide a more nuanced interpretation of findings. Lastly, the findings will be reported in accordance with open science principles, ensuring transparency in methodology, results, and interpretation.

By implementing these measures, we aim to uphold the highest standards of scientific integrity and contribute to the reproducibility and reliability of research in this field.

4.4. STATISTICAL ANALYSIS PLAN

For the confirmatory hypotheses the statistical analysis plans is as follows:

H1: We will use a chi-square goodness-of-fit test to test whether the distribution of responses on the Perceived Information Gathering Capabilities variable significantly deviates from uniform expectations.

H2: A chi-square goodness-of-fit test will be used to determine whether responses on the Perceived Invasiveness variable are evenly distributed across the three categories.

H3: We will compute the mean Acceptability Ratings for overt and covert surveillance (averaging the two items in each category) and compare them using a paired t-test.

H4: For each participant, we will calculate a difference score between their mean Acceptability Ratings for digital vs. physical surveillance. We will then use a one-way ANOVA (or Kruskal-Wallis test) to test whether this difference score varies across the three groups defined by the Perceived Invasiveness variable.

H5: For each of the four surveillance types, we will calculate a Spearman correlation between the corresponding Effectiveness Rating and Acceptability Rating. We will apply Holm-Bonferroni correction across these 4 correlations to control for multiple comparisons.

H6-8: We will compute Spearman correlations between each of the three Trust Variables (government, law enforcement, companies) and the mean of the four Acceptability Ratings. We will apply Holm-Bonferroni correction across these 3 correlations to control for multiple comparisons.

H9-11: We will compute Spearman correlations between each of the three Trust Variables and the Concern About variable (average perceived likelihood of abuse of digital and physical surveillance). We will apply Holm-Bonferroni correction across these 3 correlations to control for multiple comparisons.

H12: We will compute two Spearman correlations: One between Perceived Crime and the mean Acceptability Ratings, One between Perceived Organized Crime and the mean Acceptability Ratings. We will apply Holm-Bonferroni correction across these 2 correlations.

4.5. DATA COLLECTION

Category	Number of Participants
Total participants who started the survey	1,089
Excluded: Failed attention check	47
Excluded: Low reCAPTCHA score (< 0.5)	44
Excluded: Duration under 195 seconds	23
Excluded: No consent given	1
Excluded: Incomplete survey	44
Total excluded	107
Final included participants	982

Table 4.1.: Participant flow from initial survey entries to final sample after applying exclusion criteria.

Data collection took place over a one-week period, beginning on May 27, 2025, at 16:41 and concluding on June 3, 2025, at 16:54. A total of 1,089 participants started the survey. After applying exclusion criteria, such as failing an attention check (n=47), receiving a low reCAPTCHA score (n=44), completing the survey in less than one-third of the median time (under 195 seconds; n=23), not providing consent (n=1), or not completing the survey (n=44), a total of 107 responses were removed. An overview is presented in 4.1. This resulted in a final sample of 982 valid responses. The average time to complete the survey was approximately 11.7 minutes (701 seconds), with a median duration of 9.8 minutes (587 seconds), as illustrated in 4.2.

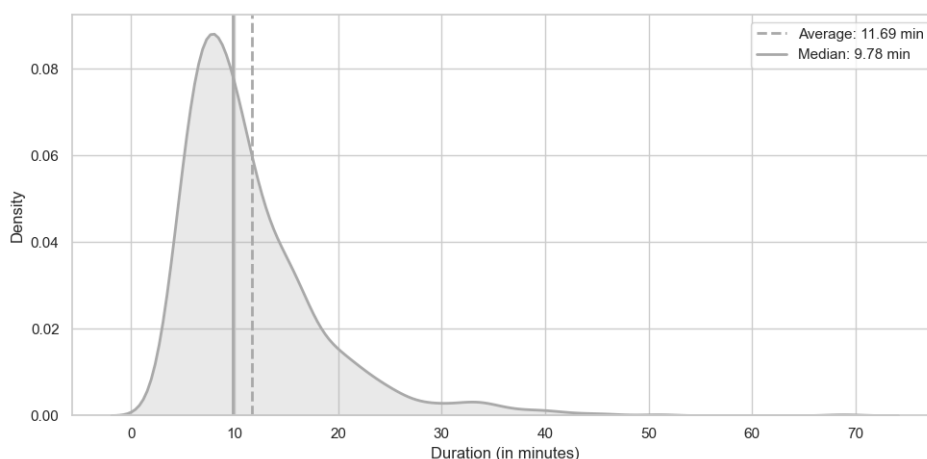


Figure 4.2.: Distribution of survey completion durations with average and median marked.

4.5.1. DATA PROCESSING AND CLEANING

Several steps were performed to prepare the dataset for analysis. First, free-text “Other” responses were manually categorized to improve consistency across demographic variables, where possible. For gender, one participant’s response was recoded as “transgender.” Nationality was grouped into three categories: U.S. American (n=962), Non-American (n=14), and Dual Citizenship (n=3), with an additional “Prefer not to say” category (n=3). Ethnicity responses were merged into meaningful categories, including White/Caucasian (n=624), Black/African American (n=135), Hispanic/Latino/a (n=62), Asian (n=54), and several mixed identities. Political affiliation responses marked as “Other” were too heterogeneous and infrequent (n=11) to form new categories and were therefore not recoded. Further details can be found in Section 5.4. For the free-text item asking which surveillance methods police should use more, common themes were grouped, and added were appropriate to existing categories. Second, questions that were presented in counterbalanced order (comparing digital and physical surveillance’s invasiveness) were recoded into a uniform variable where 1 indicated digital surveillance is more invasive, 2 indicated equal, and 3 indicated physical surveillance is more invasive. Lastly, multiple-choice questions allowing multiple selections were aggregated to create combined variables for use in quantitative analysis.

5. RESULTS

The results section is structured as follows: we begin with presenting the general results of the survey questions, followed by testing the research hypotheses. After, the results of the statistical analysis of the exploratory hypotheses are presented. Next, demographic analysis results are presented. Subsequently, there is a section on further analysis on exploratory findings, which presents a comparison of the trust responses to previous years, an analysis of the free-text answers concerning discrepancies in survey responses. Lastly, a verdict on data quality and representativeness is given.

5.1. GENERAL RESULTS OF SURVEY QUESTIONS

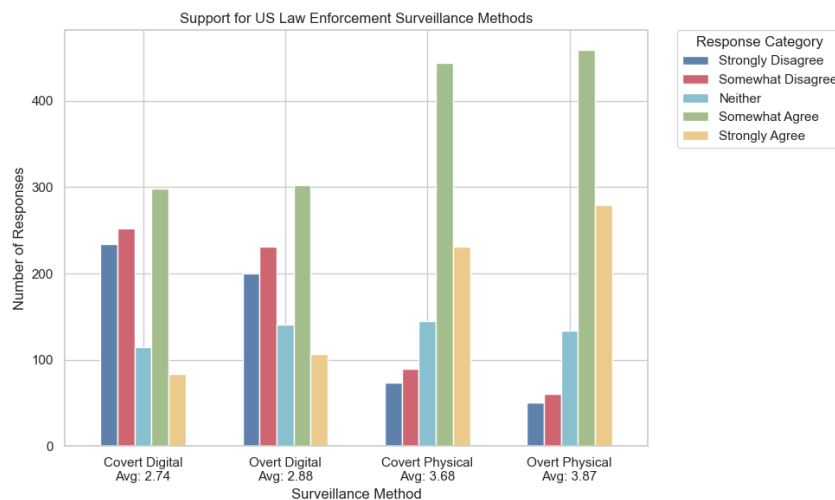


Figure 5.1.: Support for US Law Enforcement Surveillance Methods

Participants' evaluations of surveillance methods revealed clear distinctions between digital and physical forms. As shown in Figure 5.1, physical surveillance methods, particularly when overt, received stronger support than digital ones. Covert digital methods received the least support (Avg = 2.74), while overt physical surveillance was most accepted (Avg = 3.87). This pattern aligns with participants' general concerns about privacy and suggests a higher level of comfort with visible, tangible surveillance tools compared to hidden or data-driven alternatives. Interestingly, the perceived effectiveness of these methods followed a somewhat different trend. While digital methods were rated as slightly more effective overall (see Figure 5.2), the scores were more evenly distributed, indicating that participants might differentiate between what they view as effective and what they consider acceptable.

When asked about the perceived invasiveness of surveillance, digital methods were more frequently judged as invasive (29.1%) than physical ones (26.6%), though the most common response (44.3%) was that both are equally invasive. This balance in perception is reflected in participants' thinking habits as well: a majority reported occasionally (42.1%) or often (33.7%) reflecting on their digital and physical privacy, yielding an average frequency of 3.29 on a 5-point scale.

Preferences leaned mildly toward physical surveillance when participants were asked to choose between the two (31.6% vs. 24.8%), though most (43.6%) preferred neither, indicating a general hesitancy or ambivalence. In terms of perceived data-gathering capabilities, a large majority (60.9%) believed digital surveillance gathers more information, with another 30.4% saying both were equally capable. Only 8.7% credited physical surveillance with greater informational capacity.

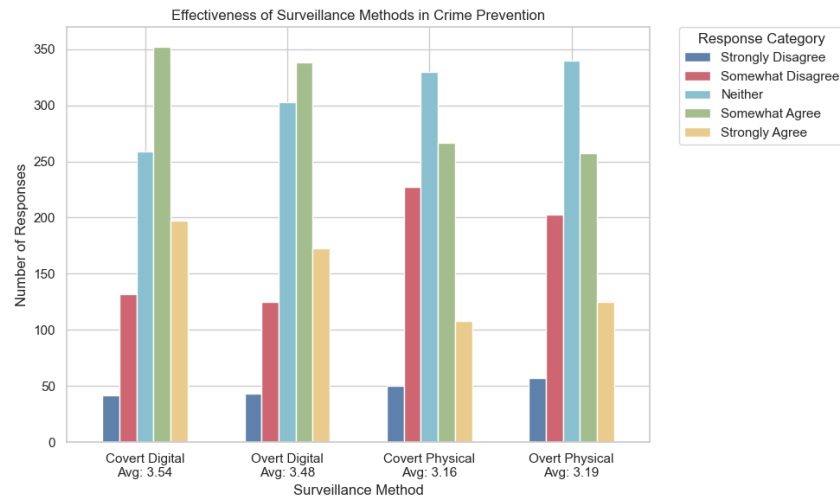


Figure 5.2.: Perceived Effectiveness of Surveillance Methods in Crime Prevention

Concerns about misuse were evident throughout. The perceived likelihood of surveillance being abused was rated as relatively high, especially for digital methods. While physical surveillance scored an average of 3.73 (on a scale from 1 to 5), digital scored 4.12. Nearly 83% of respondents saw digital abuse as somewhat or extremely likely, compared to 69% for physical methods. Refer to Plot 5.3 for details.

In terms of presence, surveillance was generally seen as prevalent in the United States. Physical surveillance averaged 6.5 on a 0–10 scale, while digital surveillance averaged 7.31. The latter was especially prominent in extreme responses, with 18% rating it a 9 and 16.6% a 10. See Plot 5.4 for details.

Despite this, personal risk was not strongly felt. The majority considered it unlikely that they would become surveillance targets within the next 12 months, with 37.9% stating “very unlikely” and 31.4% “unlikely.” The average score was 2.12. Consistent with earlier perceptions, 57.3% thought digital surveillance made them more likely targets, while only 10.1% saw physical surveillance as the greater risk.

Retrospectively, most participants believed surveillance has increased in recent years. Almost half (47.7%) said it increased significantly, and 43.8% slightly. The average rating was 4.37. Digital methods were perceived as having increased more than physical ones by a large margin (75.2%). When asked how this trend felt, nearly half (48.2%) found it worrisome, with another 36.5% stating it was “a little” worrisome.

On a broader level, government surveillance was not necessarily perceived as a threat. Most respondents felt it served to protect their freedoms (35.8%) or both protect and limit them (27.2%). Fewer viewed it primarily as a limiting force (24.8%) or neither (12.1%).

Trust across institutions was relatively low and closely spaced. Law enforcement received the lowest average trust rating (2.81), followed by private companies (2.99). Government agencies scored 3.1 in terms of legal compliance and 3.26 in doing what is morally right.

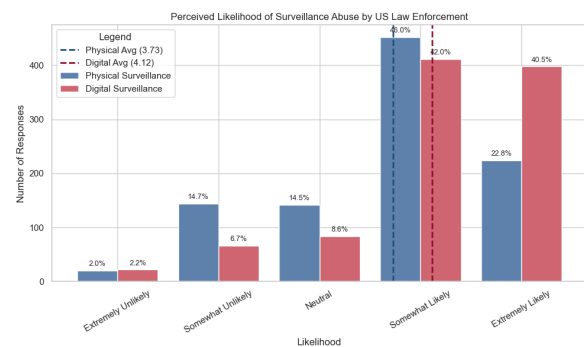


Figure 5.3.: Perceived Likelihood of Surveillance Abuse

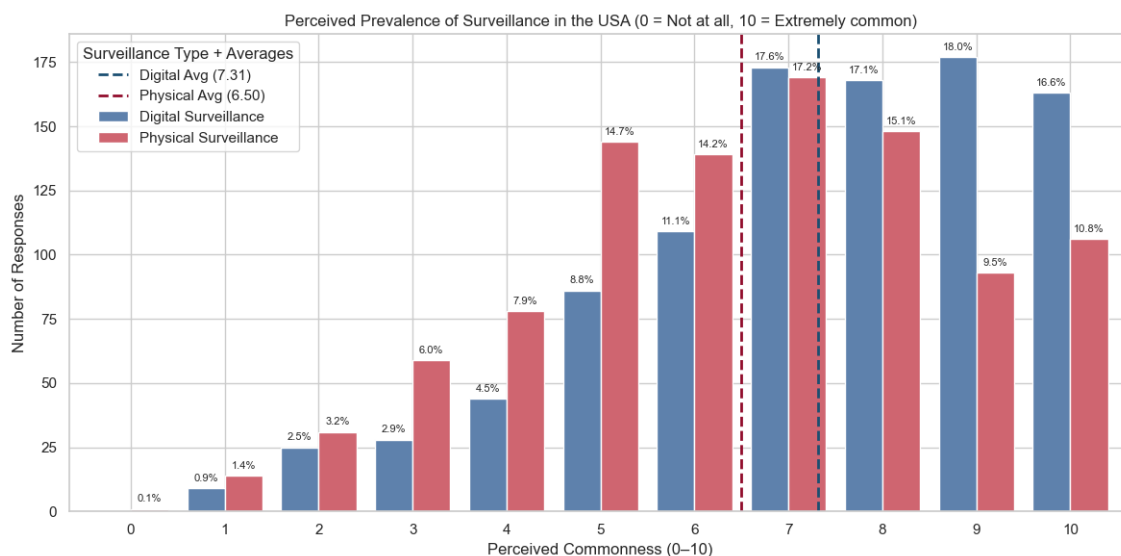


Figure 5.4.: Perceived Prevalence of Surveillance in the USA

Acceptance of specific surveillance methods revealed a more nuanced picture. Physical methods were more acceptable overall (Avg = 3.52 vs. 2.63 for digital), but responses varied by method. For instance, police body cameras (4.58) and CCTV (4.24) enjoyed high acceptance, while placing hidden cameras or GPS trackers received lower scores. Among digital methods, monitoring social media (3.2) and GPS tracking on existing devices (3.01) were more accepted than using private devices for surveillance (2.35) or installing malware (1.96). For further breakdown, see corresponding plots 5.6 and 5.5.

When asked about general societal concerns, participants saw corruption as more severe than crime or organized crime. Average severity ratings were 2.82 for corruption, 2.64 for crime and 2.29 for organized crime. See Plot 5.7 for more detail.

Respondents who viewed crime or organized crime as at least moderately severe were asked about law enforcement's surveillance access. In the case of crime, 48.9% said access was insufficient, while 29.1% said it was sufficient and 22% were unsure ($n = 499$). For organized crime, 50.1% said no, 33.7% yes and 16.2% unsure ($n = 359$). Those who answered "no" were then asked which methods law enforcement should have easier access to; the distribution is shown in the plot below 5.8.

Participants concerned about corruption were also asked whether independent third-party monitoring would increase their support for surveillance. Among the 586 who responded, 39.4% said yes, 33.4% said it wouldn't change their opinion, 24.2% said no and 2.9% were already fully supportive.

Respondents who rated both crime and corruption at least medium were asked what worried them more. Over half (55.8%) said both equally. Corruption was more concerning for 23.3%, crime for 17.7%.

Finally, the influence of political leadership was assessed. When asked if their responses would differ under a Democratic administration, most said "definitely not" (23.6%) or "probably not" (27.5%). However, 17.4% said "probably yes" and 9.7% "definitely yes." The average was 2.62 on a 5-point scale.

Those who said their answers would differ were asked how. Trust in government was expected to increase by 60.2%, decrease by 26.7% and stay the same for 13.2%. Trust in law enforcement was less affected: 42.5% said it would remain the same, 37.2% said it would increase and 20.3% decrease. Notably, 45.1% said they would fear abuse of surveillance less. Support for surveillance overall was expected to stay the same for 46.2%, increase for 33.5% and decrease for 20.3%. These responses are visualized in the plot below 5.9.

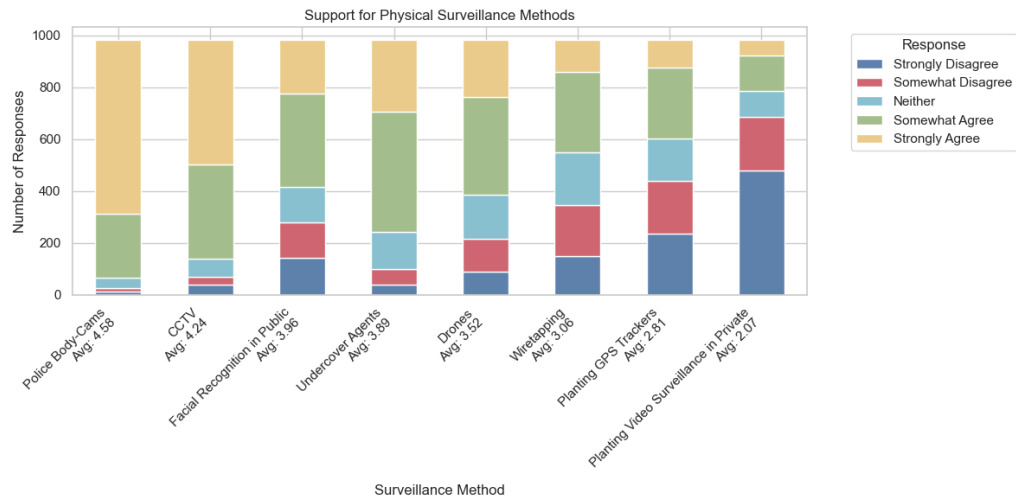


Figure 5.5.: Acceptance of specific physical surveillance methods

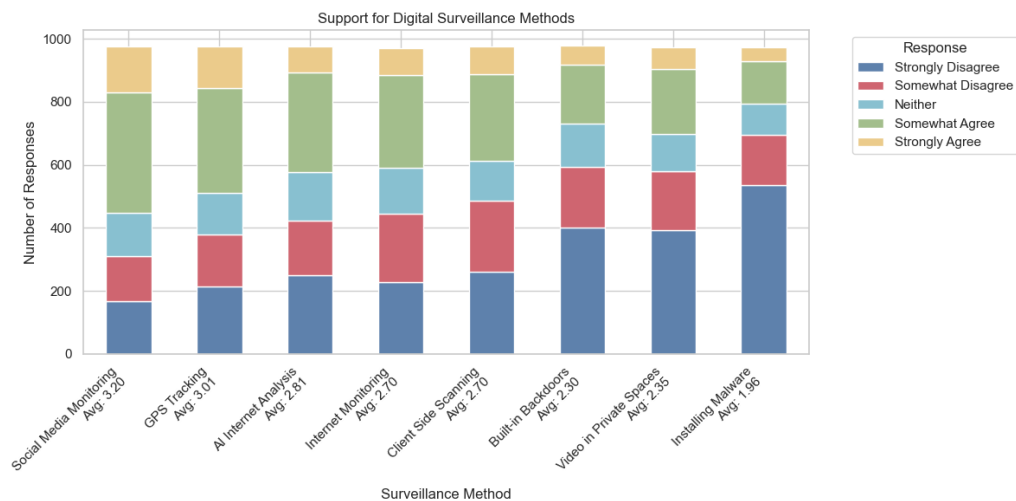


Figure 5.6.: Acceptance of specific digital surveillance methods

5.2. HYPOTHESIS TESTING I

H1 predicted that digital surveillance would be perceived as more capable of gathering information than physical surveillance. This hypothesis was supported. Among participants who saw a difference, a large majority believed digital surveillance gathers more information ($n=598$) compared to physical surveillance ($n = 85$), with fewer indicating both equally ($n = 299$). A chi-square goodness-of-fit test confirmed this distribution is highly significant ($\chi^2 = 405.67$, $p < .001$). A follow-up binomial test further confirmed that digital was significantly more often perceived as the dominant mode of information gathering ($p < .001$).

H2 investigated whether digital and physical surveillance are perceived as equally invasive. While a chi-square test indicated a significantly non-uniform distribution across the three options ($\chi^2 = 54.08$, $p < .001$), a binomial test comparing digital and physical yielded no significant result ($p = .305$). Therefore, despite some variation in responses, no clear preference for one being more invasive was found, and the hypothesis is rejected.

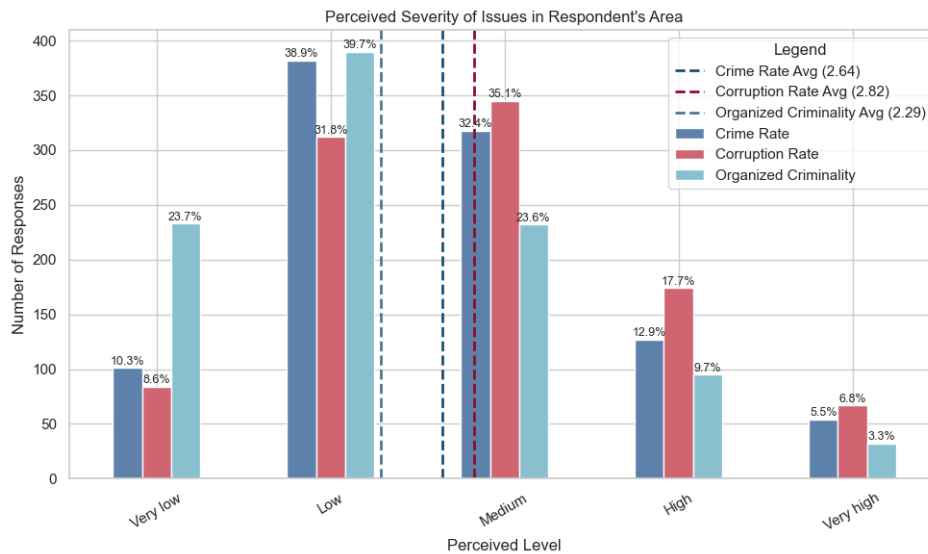


Figure 5.7.: Perceived Severity of Local Issues

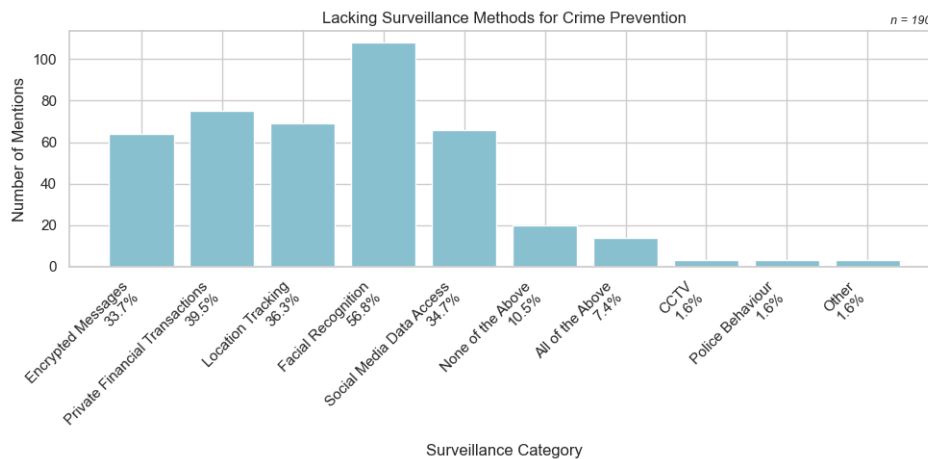


Figure 5.8.

H3 tested whether overt surveillance is perceived as more acceptable than covert surveillance. The results clearly supported this hypothesis. A paired t-test showed a substantial and statistically significant difference between the average acceptability ratings of overt ($M = 3.78$) and covert ($M = 2.81$) methods ($t = 24.28, p < .001$). Participants consistently rated overt forms as more acceptable.

H4 examined whether perceived invasiveness influences acceptance of surveillance. Participants who viewed digital surveillance as more invasive rated digital methods significantly less acceptable than physical ones (Wilcoxon $W = 6857, p < .001$), while those who found physical surveillance more invasive showed no significant difference in acceptance (Wilcoxon $W = 1947.5, p = .776$). A Kruskal-Wallis test confirmed significant differences in the digital–physical acceptability gap across groups defined by perceived invasiveness ($H = 21.05, p < .001$). This pattern was reinforced by a Spearman correlation ($\rho = 0.145, p < .001$), showing a positive association between perceived invasiveness and surveillance acceptability, particularly for digital methods. Additional Mann-Whitney U tests supported the conclusion that perceived invasiveness meaningfully shapes public acceptance, albeit with small effect sizes.

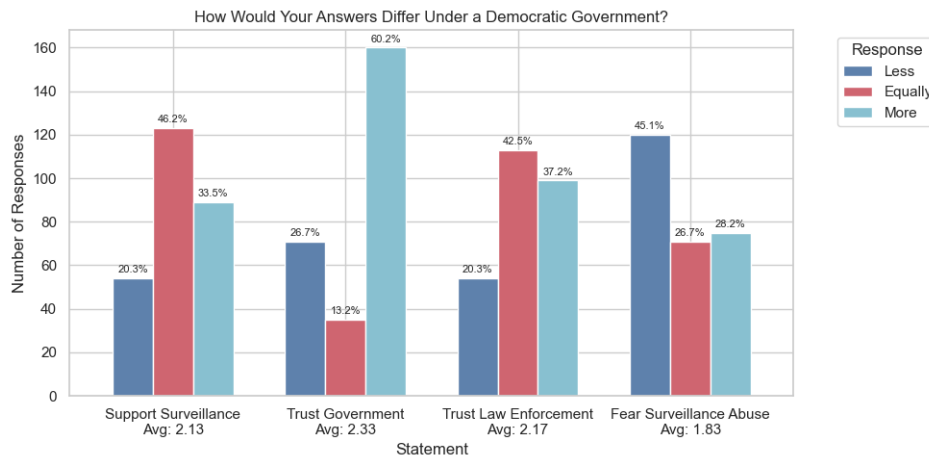


Figure 5.9.: Perceived Changes Under Democratic Leadership

H5 posited that perceived effectiveness of surveillance is positively associated with its acceptability. Spearman correlations confirmed this across all four tested types: digital covert ($\rho = 0.47$), physical covert ($\rho = 0.40$), digital overt ($\rho = 0.29$), and physical overt ($\rho = 0.30$), all significant after Holm-Bonferroni correction ($p < .001$). The strongest correlation was found for digital covert surveillance.

H6 through **H8** tested the relationship between trust in different institutions and the acceptability of surveillance. All three showed statistically significant negative correlations when controlling for multiple comparisons: trust in government agencies ($\rho = -0.35$), law enforcement ($\rho = -0.37$), and companies ($\rho = -0.24$). Note that trust was coded in reverse, so higher trust corresponds to lower numeric values. The data suggest that those who trust institutions more are more willing to accept surveillance, especially in the case of government and law enforcement.

H9 through **H11** focused on whether lower institutional trust is associated with increased concern about surveillance abuse. Strong, significant positive correlations were found for trust in government ($\rho = 0.47$), law enforcement ($\rho = 0.46$), and companies ($\rho = 0.37$). As with the previous set, reverse coding of trust scores indicates that people with lower trust in institutions report higher concern about potential misuse of surveillance.

H12 investigated whether perceptions of crime or organized crime relate to surveillance acceptance. This hypothesis was not supported. Spearman correlations showed no significant associations between perceived crime levels and acceptance of surveillance ($\rho = 0.00$ and $\rho = 0.04$, both $p > .45$). This suggests that fear of crime alone does not meaningfully shape opinions on surveillance acceptability.

Effect sizes were calculated for the first three hypotheses to provide additional insight into the magnitude of observed differences. For **H1**, which assessed the perceived capacity of different surveillance types to gather information, Cramér's V was computed from the chi-square statistic:

$$\begin{aligned}
V &= \sqrt{\frac{\chi^2}{n(k-1)}} \\
&= \sqrt{\frac{405.67}{982 \cdot (3-1)}} \\
&= \sqrt{\frac{405.67}{1964}} \approx 0.455
\end{aligned}$$

This indicates a large effect.

For **H2**, evaluating perceived invasiveness, the same formula applies with $\chi^2 = 54.08$. The resulting value is $V \approx \sqrt{\frac{54.08}{982 \cdot 2}} \approx 0.165$, corresponding to a small-to-moderate effect size.

In **H3**, a paired t-test compared the acceptability of overt and covert surveillance. Cohen's d was used to estimate the effect size, defined as $d = \frac{\bar{x}_1 - \bar{x}_2}{s_d}$, where $\bar{x}_1$ and $\bar{x}_2$ represent the group means and s_d the standard deviation of the paired differences. With a mean difference of approximately 0.97 and a pooled standard deviation of about 1.18, the effect size was $d \approx 0.82$, which is considered large.

A summary of all tested hypotheses, including statistical significance levels, effect sizes, and their corresponding magnitudes, is provided in Table 5.1. Where applicable, effect sizes are reported alongside their classification as small, medium, or large, following conventional benchmarks. The statistical method used for each hypothesis is indicated via superscript footnotes.

5.3. EXPLORATORY HYPOTHESIS TESTING II

EH1: Participants perceived digital surveillance as significantly more common ($M = 7.31$) than physical surveillance ($M = 6.50$). A Wilcoxon signed-rank test confirmed this difference was statistically significant ($p < .001$), with a medium effect size ($r = .34$). These results support the hypothesis that participants perceive differences in the perceived prevalence of digital versus physical surveillance.

EH2: A chi-square test of independence indicated a significant relationship between participants' preferred surveillance method and their perception of invasiveness ($\chi^2(4) = 282.14$, $p < .001$), with a medium-to-large effect size (Cramér's $V = .38$). Participants tended to prefer the method they perceived as less invasive. Those who perceived both as equally invasive generally reported no preference. This supports the idea that perceived invasiveness strongly influences surveillance preference.

EH3: Spearman correlation analyses revealed weak but statistically significant negative associations between the perceived likelihood of being surveilled and acceptance of surveillance: digital ($r = -0.089$, $p = .005$), physical ($r = -0.109$, $p = .001$), and overall acceptance ($r = -0.099$, $p = .002$). When broken down to method-level acceptance, a significant correlation was found only for physical methods ($r = -0.077$, $p = .016$), while digital method acceptance showed no significant association ($r = -0.010$, $p = .753$). Acceptance across all methods (combined) also showed no significant association ($r = -0.043$, $p = .180$).

EH4: A Kruskal–Wallis test revealed a significant difference in overall surveillance acceptance between participants with different beliefs about how surveillance has changed over the past five years ($H(2) = 16.78$, $p < .001$). Post-hoc Dunn tests showed participants who believed surveillance had

Table 5.1.: Summary of Hypothesis Testing Results

H	Hypothesis	Conclusion	p-value	Effect Size	Magn.
H1 ¹	Perceived differences in info-gathering capability	Accepted	<.001	$V = 0.455$	L
H2 ¹	Perceived differences in invasiveness	Rejected	.305	$V = 0.165$	S
H3 ²	Overt more acceptable than covert surveillance	Accepted	<.001	$d = 0.82$	L
H4 ^{3,4,5}	Invasiveness affects acceptability	Accepted	<.001	$\rho = .145$	S
H5 ⁵	Effectiveness → Acceptability	Accepted			
H5.1	(Digital Covert)		<.001	$\rho = .47$	M
H5.2	(Physical Covert)		<.001	$\rho = .40$	M
H5.3	(Digital Overt)		<.001	$\rho = .29$	S-M
H5.4	(Physical Overt)		<.001	$\rho = .30$	S-M
H6 ⁵	Gov trust → Acceptance	Accepted	<.001	$\rho = -.35$	M
H7 ⁵	Law trust → Acceptance	Accepted	<.001	$\rho = -.37$	M
H8 ⁵	Company trust → Acceptance	Accepted	<.001	$\rho = -.24$	S
H9 ⁵	Gov trust → Concern	Accepted	<.001	$\rho = .47$	M
H10 ⁵	Law trust → Concern	Accepted	<.001	$\rho = .46$	M
H11 ⁵	Company trust → Concern	Accepted	<.001	$\rho = .37$	S-M
H12 ⁵	Crime perception → Acceptance	Rejected	.886 / .454	$\rho = 0.00/0.04$	None

¹ Chi-square test² Paired t-test³ Wilcoxon signed-rank test⁴ Kruskal–Wallis test⁵ Spearman rank correlation

increased significantly were less accepting than those who believed it had increased slightly ($p = .006$) or stayed the same ($p = .002$). No significant difference was found between the *increased slightly* and *stayed the same* groups ($p = .250$).

EH5: Participants who believed surveillance had increased and also expressed worry reported significantly lower acceptance of surveillance compared to all other combinations of belief and concern. A Kruskal–Wallis test confirmed the group effect ($H(5) = 141.49, p < .001$), and post-hoc tests showed that this group differed significantly from all others. No other pairwise comparisons were significant. This indicates that concern only influences acceptance when combined with the perception of increasing surveillance.

EH6: A chi-square test showed a significant association between perceived changes in surveillance and worry about those changes ($\chi^2(8) = 97.54, p < .001$). Among those who believed surveillance had increased significantly, 63% reported being worried. In contrast, only 23–25% of those who believed surveillance had stayed the same or decreased slightly were worried. The "not worried" and "a little worried" categories were most frequent among those who did not perceive an increase.

EH7: Participants who reported stronger feelings of being disadvantaged by the U.S. government or law enforcement were significantly more likely to report lower trust and greater concern about abuse. For perceived disadvantage by the government: trust in government ($r = -0.438, p < .001$) and concern

about abuse ($r = -0.331, p < .001$). For law enforcement: trust ($r = -0.389, p < .001$), concern ($r = -0.282, p < .001$). All associations were moderate in size.

EH8: The issue participants were most concerned about (crime vs. corruption vs. both vs. neither) was associated with surveillance acceptance ($H(3) = 16.64, p < .001$). Dunn's post-hoc comparisons showed that participants who were primarily concerned with crime reported significantly higher acceptance than those concerned with corruption ($p = .009$). No other pairwise differences reached significance.

EH9: Perceived levels of corruption were associated with lower institutional trust. Spearman correlations showed a small but significant association between higher perceived corruption and lower trust in government ($r = 0.193, p < .001$) and law enforcement ($r = 0.204, p < .001$).

EH10: A chi-square test revealed no significant difference in support for independent monitoring among participants who perceived high corruption and either supported or opposed surveillance ($\chi^2(4) = 5.61, p = .132$). This indicates that perceived corruption alone does not predict a preference for oversight mechanisms.

EH11: 9.3% of participants ($n = 91$) displayed a pattern of general disagreement with a surveillance type but supported most specific methods (Q128), while 16.3% ($n = 160$) showed the opposite pattern. This suggests some participants apply inconsistent reasoning between general and specific attitudes toward surveillance.

EH12: Of those who indicated which method they thought gathered more information ($n = 163$), 85% selected digital surveillance. Of these, 10.1% ($n = 99$) also identified that method as more invasive. About 9.5% held mismatched views (e.g., digital gathers more, but physical is more invasive). Additionally, 8.9% preferred the method they also found more invasive.

EH13–14: Demographic analyses (see Section 5.4 for further details) revealed that age, income, and education were significantly associated with several variables, including trust, acceptance, and perceived effectiveness. Although the relationships varied and effect sizes were small, the results support the hypothesis that these demographic factors shape perceptions of surveillance. Age was associated with trust in government, and overall effectiveness, income with all of the mentioned variables and education with all except overall perceived effectiveness.

EH15: Participants politically aligned with the governing party (Republican) showed higher surveillance acceptance ($M = 3.51$ vs. 3.17), lower concern ($M = 3.65$ vs. 4.10), and lower trust in government ($M = 2.80$ vs. 3.41), all $p < .001$. These results suggest that political alignment significantly affects surveillance attitudes. Further details are in Section 5.4.

EH16: Attitudes toward how responses would change under a Democratic government differed significantly by party affiliation across four dimensions (support, trust in government, trust in law enforcement, and concern about abuse), all $p < .001$. Democrats generally reported increased trust and reduced fear, while Republicans reported the opposite. Independent participants showed more mixed but still patterned responses. Free-text responses confirmed that partisan alignment plays a central role in shaping surveillance attitudes and perceived legitimacy of authority and that it is a critical factor for many participants.

The free-text responses reveal highly polarized and emotionally charged attitudes toward surveillance and government trust, deeply intertwined with political affiliations. Many participants express a strong sense of distrust toward the current U.S. administration, particularly under Republican leadership and

specifically Donald Trump. These respondents often articulate fear of political weaponization of surveillance, referencing scenarios such as being targeted for dissent, sent to prison without due process, or facing discrimination based on race or political views. The tone in these responses is frequently intense, often marked by anger, anxiety, and a sense of urgency. Language is at times harsh or vulgar, with several individuals swearing or using extreme labels to emphasize their positions.

In contrast, others display deep skepticism toward Democrats, accusing them of abusing institutional power, promoting authoritarian policies, or engaging in widespread corruption. Some responses frame the Democratic party as morally or ideologically dangerous, referring to them as “Marxists,” “reactionary,” or “perverted.”

At the same time, a number of respondents suggest that, while surveillance itself is problematic, it would be more acceptable or less concerning under a Democratic administration due to a perceived greater emphasis on civil liberties, legal oversight, and institutional integrity. Among these participants, trust in government is conditional on leadership and regulatory transparency. They indicate that the presence of checks and balances, such as court orders or warrants, would increase their comfort with surveillance measures.

Other participants take a more resigned or system-critical stance, expressing that surveillance abuse is not limited to one party but reflects a broader erosion of democratic protections and digital rights. These views are often accompanied by a more reflective tone, focusing on long-term structural issues like the entanglement of corporate and state surveillance or the weakening of due process.

Overall, the tone ranges from fearful and paranoid to cynical, outraged, and occasionally hopeful. There is little middle ground. Those who support Democrats tend to associate them with stronger ethics, concern for human rights, and lawful conduct, whereas those who oppose them view the party as equally or even more dangerous than their Republican counterparts. These narratives underscore the intense affective polarization in the U.S., particularly in how citizens perceive and evaluate the legitimacy and purpose of surveillance.

5.4. DEMOGRAPHIC ANALYSIS

To explore how individual differences shape attitudes toward surveillance, a series of bivariate analyses was conducted between eleven dependent variables and seven demographic predictors. Depending on the data types, Spearman’s rank correlation coefficients were computed for ordinal and continuous pairings, while Mann-Whitney U, Kruskal-Wallis H, and chi-square tests were applied for categorical group comparisons. For non-correlation tests, raw statistics were normalized to a 0–1 scale in order to provide a visual comparison of effect strength in the corresponding heatmap 5.10.

It visualizes all statistically significant relationships with uncorrected p-values below .05. Each cell reflects both the magnitude of the effect and where applicable the direction, ranging from deep blue/green (strong negative association) through yellow (no association) to deep red (strong positive association).

To control for multiple comparisons, two standard correction procedures were applied: Holm-Bonferroni and Benjamini-Hochberg (FDR). A full overview of uncorrected and corrected significance is presented in Table 5.3, where results are colour-coded by the number of significance thresholds passed. Green cells indicate that the result remained significant under all three thresholds, orange denotes significance under uncorrected and FDR but not Holm correction, and red indicates results that only met the uncorrected significance criterion.

Political orientation consistently emerged as the most influential demographic predictor. It was significantly associated with nearly all dependent variables, and these associations remained robust after correction across all thresholds. Specifically, political orientation was strongly associated with acceptance of both digital and physical surveillance methods, concern about potential surveillance abuse, trust in government, law enforcement, and companies, as well as the perceived effectiveness and purpose of surveillance.



Figure 5.10.: Overview of Demographic Analysis (Effect Sizes Normalized for gender and Political Affiliation)

Table 5.2.: Exploratory Hypothesis Test Summary (Condensed)

EH	Topic		p	Effect	Mag.
EH1	Digital > Physical commonness	1	< .001	$r = .34$	M
EH2	Invasive $\leftrightarrow$ Preference	1	< .001	$V = .38$	M-L
EH3a	Likelihood $\rightarrow$ Accept (Digital)	1	.005	$\rho = -.09$	S
EH3b	Likelihood $\rightarrow$ Accept (Physical)	1	.001	$\rho = -.11$	S
EH3c	Likelihood $\rightarrow$ Accept (Overall)	1	.002	$\rho = -.10$	S
EH4	Trend $\rightarrow$ Acceptance	1	< .001	$\hat{\epsilon}^2 \approx .02$	S
EH5	Trend $\times$ Worry $\rightarrow$ Acceptance	1	< .001	$\hat{\epsilon}^2 \approx .14$	M
EH6	Trend $\rightarrow$ Worry	1	< .001	$V = .23$	S-M
EH7	Disadvantage $\rightarrow$ Trust / Concern	1	< .001	$\rho = -.28$ to $-.44$	M
EH8	Focus (Crime vs. Corr.) $\rightarrow$ Accept	1	< .001	$\hat{\epsilon}^2 \approx .02$	S
EH9a	Corr. $\rightarrow$ Trust Gov	1	< .001	$\rho = .19$	S
EH9b	Corr. $\rightarrow$ Trust Law	1	< .001	$\rho = .20$	S
EH10	Monitoring preference (low trust)	0	.132	—	—
EH11	General/method inconsistency	—	—	—	—
EH12	Info-driven method preference	—	—	—	—
EH13	Demogr. $\rightarrow$ Accept / Trust / Invasiveness	1	< .05	$\rho \approx .10$ –.20	S
EH14	Educ./Income $\rightarrow$ Digital Accept	1	< .05	$\rho \approx .10$ –.20	S
EH15	Political align. $\rightarrow$ Accept / Trust / Fear	1	< .001	$\hat{\epsilon}^2 \approx .08$	M
EH16	Party modulates Democrat scenario	1	< .001	$V = .25$ –.35	M

1 = Accepted, 0 = Rejected; — = Descriptive only.

S = Small, M = Medium, L = Large, M-L = Medium-Large, S-M = Small-Medium.

Tests: Wilcoxon (EH1), Chi-square (EH2, EH6, EH10, EH16), Spearman (EH3, EH7, EH9, EH13–14), Kruskal-Wallis or Mann-Whitney (EH4–5, EH8, EH15).

Income shows influence across the most dimensions, however with rather weak effects. It is associated with higher acceptance of both digital and physical surveillance methods, greater perceived overall effectiveness. These effects, although not as strong as those of political orientation, point to a trend in which individuals with higher income levels may view surveillance more favorably or as less threatening. Education also plays a role, particularly in shaping acceptance and perceptions of purpose, though its effects remain modest. Interestingly, education is positively associated with acceptance of digital surveillance and the frequency of privacy-related thoughts, but negatively associated with trust in institutions.

Income and education were also found to be meaningful predictors across many variables. Higher income levels were associated with increased acceptance of surveillance and greater perceived effectiveness. Education showed a more nuanced pattern. On one hand, higher educational attainment was positively associated with perceived surveillance effectiveness and privacy-related reflection. On the other hand, education was negatively associated with trust in government and law enforcement, potentially reflecting a more critical stance toward authority among highly educated individuals. Not all of these effects remained significant after Holm correction, but several did persist under the FDR procedure. Age was significantly associated with multiple outcomes in the uncorrected analysis, including lower acceptance of digital surveillance, more frequent privacy-related thoughts, and higher ratings of surveillance effectiveness. However, only the associations with privacy thoughts and effectiveness remained significant after correction. These findings suggest that younger individuals may be more reflective or critical of surveillance practices, although the effects were generally modest.

DV	IV	Uncorr.	Holm	BH
Acceptance of Digital Methods	Age	1	1	1
Acceptance of Digital Methods	Education	1	1	1
Acceptance of Digital Methods	Income	1	1	1
Acceptance of Digital Methods	Political	1	1	1
Acceptance of Physical Methods	Education	1	1	1
Acceptance of Physical Methods	Income	1	1	1
Acceptance of Physical Methods	Political	1	1	1
Concern About Surveillance Abuse	Education	1	0	0
Concern About Surveillance Abuse	Income	1	1	1
Concern About Surveillance Abuse	Political	1	1	1
Frequency of Privacy-Related Thoughts	Age	1	1	1
Frequency of Privacy-Related Thoughts	Ethnicity	1	0	1
Frequency of Privacy-Related Thoughts	Education	1	1	1
Frequency of Privacy-Related Thoughts	Income	1	1	1
Frequency of Privacy-Related Thoughts	Tech Affinity	1	1	1
Overall Acceptance	Age	1	0	0
Overall Acceptance	Income	1	1	1
Overall Acceptance	Political	1	1	1
Perceived Likelihood of Being Surveilled	Age	1	0	1
Perceived Overall Effectiveness	Age	1	1	1
Perceived Overall Effectiveness	Ethnicity	1	0	1
Perceived Overall Effectiveness	Income	1	1	1
Perceived Overall Effectiveness	Political	1	1	1
Perceived Purpose of Surveillance	Age	1	0	1
Perceived Purpose of Surveillance	Gender	1	1	1
Perceived Purpose of Surveillance	Education	1	1	1
Perceived Purpose of Surveillance	Income	1	1	1
Perceived Purpose of Surveillance	Political	1	1	1
Trust in Companies	Age	1	0	0
Trust in Companies	Ethnicity	1	1	1
Trust in Companies	Education	1	1	1
Trust in Companies	Income	1	1	1
Trust in Companies	Political	1	1	1
Trust in Companies	Tech Affinity	1	0	0
Trust in Government	Age	1	1	1
Trust in Government	Ethnicity	1	1	1
Trust in Government	Education	1	1	1
Trust in Government	Income	1	1	1
Trust in Government	Political	1	1	1
Trust in Law Enforcement	Ethnicity	1	0	1
Trust in Law Enforcement	Education	1	1	1
Trust in Law Enforcement	Income	1	1	1
Trust in Law Enforcement	Political	1	1	1

Table 5.3.: Significance of demographic predictors before and after correction. Green = significant across all thresholds, orange = uncorrected + one correction, red = only uncorrected.

Ethnicity explained significant variation in trust-related and perceptual outcomes. Group differences were observed for trust in companies and government, as well as perceived surveillance effectiveness. However, several of these effects did not survive correction for multiple testing, suggesting that the role of ethnicity in shaping surveillance attitudes may be more context-dependent or subject to variability within groups.

Gender and self-reported tech affinity yielded the fewest significant associations, and most did not remain significant after correction. Although isolated uncorrected effects were observed (such as tech affinity predicting concern about abuse and privacy-related thoughts) these did not survive Holm correction and should be interpreted with caution. Gender was not significantly associated with any dependent variable after correction.

Taken together, these findings highlight the differential role of demographic variables in predicting public perceptions of surveillance. Political orientation consistently accounts for significant variance across acceptance, concern, and trust-related measures. Income and education contribute additional explanatory power, whereas age, ethnicity, gender, and tech affinity appear to exert more selective or indirect influence. These patterns underscore the importance of incorporating demographic context when analyzing support for and resistance to surveillance practices.

A more detailed overview is presented in Figure 5.10. The deeper the colour, the larger the effect-size, normalized to a range from 0-1.

The results presented above were not corrected for multiple comparisons. Given the large number of statistical tests conducted, this increases the risk of Type I errors (false positives). While the uncorrected p-values provide an initial overview of potential associations, caution is warranted when interpreting marginally significant results. Applying corrections for multiple testing as the Holm-Bonferroni method for family-wise error control or the Benjamini-Hochberg procedure for controlling the false discovery rate would reduce the number of statistically significant findings.

5.5. FURTHER ANALYSIS AND EXPLORATORY FINDINGS

5.5.1. COMPARISON OF TRUST RESPONSES

To contextualize the trust results, comparisons were drawn with findings from Geierhaas et al., who conducted a similar U.S.-based survey on institutional trust between December 15, 2023, and January 31, 2024 [13]. The four questions on trust were explicitly identical, to allow for proper comparison. Sample sizes were also similar: Geierhaas et. al having a sample size of 977, this study having a sample size of 982. While their study found relatively high levels of trust: 58.5% in law enforcement, 50.8% in companies, 48.7% in government agencies, and 41.8% in the government to do the right thing, our sample reported substantially lower agreement: 48.1%, 43.8%, 37.7%, and 31.4% respectively.

Two-sample z-tests for proportions confirmed that these differences are statistically significant across all four entities. Agreement with the statement that the government can be trusted to do the right thing was significantly lower in the present study than in Geierhaas et al. ($z = -4.78$, $p < .001$), as was trust in government agencies ($z = -4.93$, $p < .001$), private companies ($z = -3.09$, $p = .002$), and law enforcement ($z = -4.65$, $p < .001$).

5.5.2. FREE-TEXT ANALYSIS DISCREPANCIES

There were three conditional questions that were asked when certain conditions were met. The first was if someone generally disagreed with one of the four surveillance types (digital overt, digital covert, physical overt, physical covert) but then agreed with all or most of the corresponding specific surveillance methods later in the study. The reverse was also a separate question, where participants showed general agreement, but disagreed with all or most corresponding methods. The last conditional question was triggered if participants would prefer the method for their personal surveillance that they also believe is capable of gathering more information. The questions were then free-text answers where the participants were asked

Table 5.4.: Agreement Rates with Trust Statements Across Years

Statement	2023 (%)	Jan 2024 (%)	2025 (%)
In general, I trust the government to do the right thing.	–	41.8	31.4
In general, I trust government agencies to comply with the law.	–	48.7	37.7
In general, I trust companies such as Google and Apple to comply with the law.	–	50.8	43.8
In general, I trust that law enforcement officials will treat me fairly.	–	58.5	48.1

for the reasons behind their choices.

Of those participants who expressed general disagreement with a surveillance type but nonetheless supported specific methods, this was often with the background that they perceived these as justified, effective, or minimally invasive. It was often a more nuanced, conditional stance rather than inconsistency.

A key theme frequently mentioned was contextual justification. Many respondents acknowledged that while they opposed for example covert surveillance in principle, certain methods were acceptable if used in serious cases. These participants emphasized that tools like undercover agents, drones, or digital tracing techniques could be necessary in high-stakes contexts, especially when conducted with proper legal oversight (e.g. warrants, court approval).

Some individuals described a change in perspective over the course of the survey, with one respondent noting that as they progressed and reflected more deeply on the specific methods, they came to realize that law enforcement should be allowed to use some of them under the right circumstances. This indicates how increased exposure to nuanced scenarios can shift perceptions from abstract rejection to conditional acceptance.

Crime prevention was a major reason participants cited for making exceptions. Several noted that although they were uneasy with the idea of especially covert surveillance broadly, specific tools were effective in fighting crime or gathering evidence. In these cases, pragmatism overrode abstract opposition. Additionally, it was mentioned numerous times that these surveillance methods are not necessarily just for gathering evidence against criminals, but also in order to protect victims or civilians. An example mentioned on several occasions was for example using location tracking to find a missing person or in case of emergency.

Some people differentiated between methods based on perceived invasiveness. For example, bodycams, CCTV, or facial recognition in public spaces were considered relatively benign, especially when compared to more controversial tactics like client-side scanning or deep digital monitoring, which were more likely to be rejected even by those open to surveillance in general.

A number of responses emphasized the importance of legal clarity and ethical constraints. While general covert surveillance evoked concerns about abuse and intrusion, participants were more open to specific methods if they followed clear rules, targeted only suspects, and excluded private spaces, the way several of the methods were framed in the survey (like CCTV in public spaces).

For those who expressed overall agreement with surveillance in principle but rejected most corresponding specific methods when prompted to evaluate them individually, this was often rooted not in contradiction, but in careful ethical reasoning, legal concerns, and fear of overreach.

For many, the idea of particularly digital surveillance for public safety or crime prevention was acceptable in theory, especially when framed as targeted or warranted. However, once presented with individual techniques like client-side scanning, AI analysis, or installing malware, participants raised

significant objections. These methods were widely perceived as too invasive, lacking transparency, and prone to misuse, bias, or false positives. Even supporters of surveillance acknowledged that such tools, if unchecked, could violate constitutional rights and chill free speech or autonomy.

A recurring concern was the lack of legal safeguards. Many participants stressed that surveillance should only occur with proper judicial oversight, such as a warrant, and should be directed at suspected criminals, not applied indiscriminately to the general population. Some explicitly referenced the Fourth Amendment or expressed distrust in law enforcement's ability to ethically handle such powerful tools without abusing them.

Another key theme was the disconnect between general acceptance and emotional response to specific methods. As participants dug deeper into the survey, some described how methods that initially seemed acceptable became deeply unsettling once they imagined them applied to themselves. This illustrates a gap between abstract support and practical discomfort, especially when methods involved surveillance in private spaces, predictive algorithms, or covert access to personal devices.

Others mentioned that even if they weren't doing anything wrong, the principle of being watched without consent or knowledge felt dystopian.

For the final question, despite acknowledging that digital surveillance may reveal more personal information, many participants expressed a clear preference for it over physical surveillance due to its perceived psychological distance, controllability, and reduced sense of violation. Physical intrusion was commonly described as creepy, emotionally distressing, and an invasion of immediate personal space. In contrast, digital surveillance was often seen as abstract, passive, or "in the background," making it feel less invasive. The phrase "out of sight, out of mind" reflected a widely shared view: participants found digital tracking tolerable simply because it was less visible or felt less immediate.

Additionally, it was mentioned that digital surveillance often felt "inevitable" and was happening anyways. Their digital footprint was already being tracked by companies, so government surveillance didn't feel like a significant additional intrusion. On the contrary, several participants emphasized a sense of control over digital exposure, noting that they could regulate their online presence, use encryption or VPNs, or were at least aware that everything online could be observed. Others again described themselves as "boring online" or as deliberately cautious about what they share, reducing concern about being monitored.

The argument "I have nothing to hide" appeared frequently, often accompanied by the belief that surveillance would only be used in emergencies or for public safety—scenarios in which digital methods were seen as faster and more effective. Because most daily activity now occurs online, participants felt that digital surveillance would yield richer, more targeted insights if needed.

However, a smaller group preferred physical surveillance precisely because they saw it as more visible, concrete, and legally bound. They viewed the need for warrants or physical access as meaningful safeguards, making physical methods more transparent and less prone to silent abuse than digital data collection.

In sum, preferences were shaped less by how much information could be gathered and more by perceived control, legal oversight, emotional comfort, and normalization of data tracking.

5.6. VERDICT ON DATA QUALITY AND REPRESENTATIVENESS

The final dataset consists of 982 usable responses, which is sufficient for the statistical analyses conducted in this study. By design, the dataset contains no missing values for any of the analyzed variables. The survey was structured to enforce completion of all relevant fields, ensuring that the data is both complete and clean. Data collection was carried out over a concise and consistent period, further contributing to internal validity and minimizing the risk of temporal bias due to external events.

With respect to representativeness, the sample offers reasonably broad demographic coverage of the U.S. adult population and can be considered directionally representative, though not fully balanced across

all categories. Since only participants aged 18 and older were eligible, demographic comparisons were made against U.S. Census estimates for the adult population.

Gender representation closely aligns with national distributions. Age distribution shows moderate skew toward older age groups. Educational attainment and household income span a wide spectrum but are tilted slightly toward higher education levels and middle to upper income brackets. Ethnic diversity is present, although Black and Hispanic Americans are somewhat underrepresented relative to their national proportions.

Despite these imbalances, the sample provides a robust foundation for identifying general trends and public attitudes. Interpretations of the results should be made with awareness of these demographic characteristics. For enhanced generalizability, post-stratification weighting or stratified subgroup analysis may be considered in future work.

A more detailed comparison between the survey sample and national benchmarks is provided in the following figures.

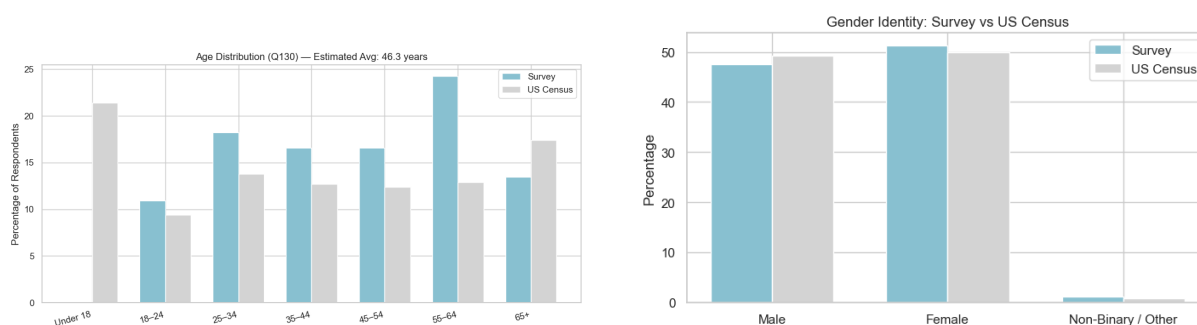


Figure 5.11.: Comparison of Sample and U.S. Population: Age and Gender

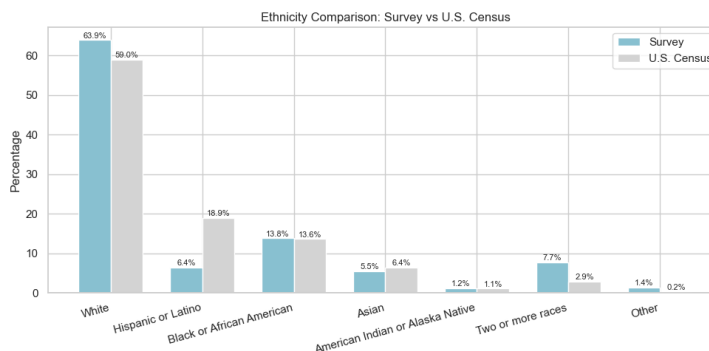


Figure 5.12.: Comparison of Sample and U.S. Population: Ethnicity

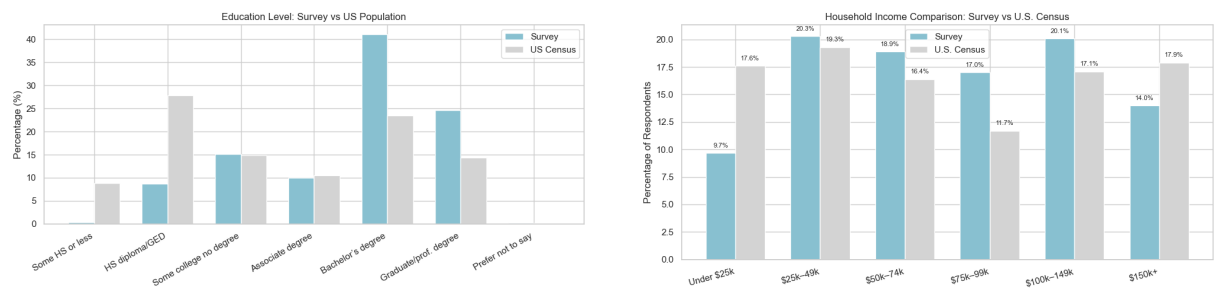


Figure 5.13.: Comparison of Sample and U.S. Population: Education Level and Household Income

6. DISCUSSION

The findings of this study offer a nuanced perspective on public attitudes toward surveillance in the United States, both in digital and physical contexts. Overall, participants showed a clear preference for overt surveillance methods over covert ones. This aligns with previous literature [3] suggesting that transparency and awareness mitigate concerns of abuse and foster greater public acceptance. Surprisingly, physical surveillance methods were rated slightly more acceptable than digital ones, which contradicts the intuitive assumption that digital methods, due to their more passive and less visible nature, might be perceived as less intrusive.

In terms of perceived effectiveness, overt surveillance was consistently rated as more effective than covert surveillance. This may be interpreted as a reflection of the “Big Brother is watching” effect, where awareness of being monitored acts as a deterrent. Digital surveillance, in turn, was also judged as slightly more effective than physical methods. Importantly, effectiveness played a central role in shaping acceptance: the more effective a method was perceived to be, the more acceptable it became, regardless of perceived invasiveness. This reflects an instrumental logic among participants, suggesting that when surveillance is seen as genuinely useful in ensuring safety or deterring crime, concerns about intrusiveness may be secondary.

The role of invasiveness, however, was more complex. Most participants rated digital and physical surveillance as equally invasive, and no surveillance type stood out as significantly more invasive across the sample. Yet, a fascinating order effect was uncovered: participants disproportionately selected the option that was presented first (e.g., “Digital surveillance is more invasive than physical” vs. the reverse), suggesting that the question framing had more influence than actual content. This finding suggests there are blurring boundaries between digital and physical surveillance in the public consciousness. Participants may not have a clear conceptual distinction between the two, especially as surveillance ecosystems increasingly merge online and offline modalities (e.g., AI-enhanced CCTV, location-based tracking, and data fusion systems).

While invasiveness was negatively correlated with acceptance, the effect size was small, further reinforcing that perceived effectiveness had greater explanatory power for acceptability judgments. However, invasiveness did significantly influence individual preferences, with participants typically favoring the type they found less invasive. When both were perceived as equally invasive, they were also rated as equally acceptable, indicating a consistent, if moderate, influence on decision-making.

A key psychological insight lies in participants’ perception of surveillance prevalence. While most respondents believed that digital surveillance is widespread and likely to be abused, only 15% felt personally at risk of being surveilled. This gap between concern at a societal level and a sense of personal immunity mirrors a broader trend. A similar pattern can be seen in what is referred to in privacy research known as the “privacy paradox.” This refers to studies revealing that for example while privacy is stated as a primary concern for citizens, individuals are quick and willing to reveal personal information about themselves[22]. The perceived likelihood of being surveilled was negatively associated with acceptance, but the effect size was extremely weak ($r < 0.1$), and only slightly stronger for physical methods. This suggests that personal threat perception plays only a minimal role in shaping general attitudes, compared to perceived societal benefits.

Participants also overwhelmingly believed that surveillance has increased over the past five years, and this trend was viewed as troubling. This finding reinforces that surveillance is a highly salient issue, particularly in an age marked by growing technological capabilities and ongoing geopolitical tensions.

Interestingly, when asked to prioritize surveillance in response to societal issues, corruption was rated as a more serious concern than crime, including organized crime. However, when examining those who rated both corruption and crime as medium to high concerns, a split emerged: those who considered crime

the more pressing issue were more accepting of surveillance, while those who ranked corruption higher were less accepting. This highlights a critical tension: while some view surveillance as a necessary tool for ensuring public safety, others fear it may be misused by corrupt authorities, thus eroding trust and civil liberties.

Participants who viewed corruption as a key problem also expressed significantly lower trust in authorities, with a small but meaningful correlation to lower acceptance of surveillance overall. Trust in government and law enforcement were both strongly associated with reduced fear of surveillance abuse ($r \approx -0.46$ to -0.47), while trust in companies was slightly lower ($r \approx -0.35$). This suggests that the public sees a direct link between state actors and corporate enablers of surveillance, a theme echoed in several open-ended responses.

The qualitative responses revealed deeper reasoning and emotional responses behind these patterns. A recurring theme was the idea that “out of sight, out of mind” makes physical surveillance more disturbing. People are unsettled by the idea of being watched by a human or device in real time, even if they accept digital monitoring as inevitable. Many respondents drew comparisons between corporate data tracking and government surveillance, suggesting a resignation to being watched online and arguing that, if companies are already collecting such data, it would make sense for the government to do so too provided it’s done transparently and for security purposes.

Another major insight was the belief that some surveillance methods should only be used in emergencies or under strict legal oversight, such as search warrants or court orders. A notable number of participants referenced the Fourth Amendment and the importance of clear legal safeguards. This legal framing was especially prominent among those who supported physical surveillance despite finding it more invasive, arguing that physical methods are easier to regulate and less prone to silent abuse.

The narrative split between general attitudes and method-specific judgments was also prominent. Many participants who generally supported surveillance rejected most individual methods, citing them as too invasive, technologically overpowered, or prone to misuse. Several mentioned that while the idea of overt digital surveillance seems acceptable, the reality (AI scanning, client-side scanning, or social media monitoring) was uncomfortable. Client-side scanning was one of the most frequently criticized tools, with participants expressing concerns about its potential for deep intrusion into private digital spaces.

Conversely, some participants who were generally against surveillance expressed support for specific techniques they felt were necessary to fight crime, such as GPS tracking or CCTV. This group often emphasized conditional acceptance that under clear, targeted, and legally approved circumstances, certain surveillance tools were justified.

An especially compelling theme came from participants whose opinions shifted over the course of the survey. One respondent shared that, as the study progressed, they began to reflect more deeply on the topic and recognized the value of surveillance “under the right conditions.” This moment of realization captures the internal tension many participants faced, balancing personal privacy with public safety, and navigating complex trade-offs in a digital society.

Finally, the demographic and political analyses further illustrated the divisiveness of the issue. Political affiliation emerged as one of the strongest predictors of surveillance attitudes. Interestingly, however, the demographic effects were limited in magnitude. Only age showed a modest effect on perceived surveillance likelihood, while income and education influenced some acceptance patterns but not others. These findings may reflect the highly polarized nature of the surveillance debate, especially in the U.S. context.

Taken together, these results demonstrate that while general trends in surveillance acceptance are observable, attitudes are often method-specific, context-dependent, and emotionally charged. Participants weigh effectiveness more heavily than invasiveness, but their views are filtered through trust, personal beliefs, political identity, and legal frameworks. As surveillance technologies continue to evolve, public discourse must engage more deeply with these layered perceptions and policymakers must ensure that the

tools employed are not only effective, but also transparent, accountable, and deserving of the public's trust.

6.1. LIMITATIONS

Several limitations should be considered when interpreting the findings of this study. First, not all aspects of the survey design were ideal. A planned experimental component regarding question order effects could not be analyzed. The survey was designed to examine whether answering a general acceptance question before or after questions on specific surveillance methods would influence responses. Although Qualtrics randomly presented the two question sets in different orders, identifying which participant received which sequence required an additional embedded tracking variable. This variable was not implemented before data collection began, and as a result, the order of presentation could not be reconstructed from the exported data. Consequently, the planned analysis of order effects was not possible, and any potential influence of question sequencing remains untested. Additionally, the survey instruments did not explicitly mention whether surveillance actions would occur with or without a warrant (or other legal authorization such as court orders or subpoenas). This omission matters because willingness to accept surveillance often hinges on perceived legality and oversight e.g. the presence of judicial authorization, the stated purpose (criminal investigation vs. national security), the evidentiary threshold, and the expected scope and duration of data access. Without clarifying the legal context, respondents may have answered based on differing assumptions about due process and safeguards, which could attenuate or inflate measured acceptance and concern. Future iterations should specify the legal basis (e.g. "with a warrant," "emergency exception," or "no warrant") and, where relevant, the issuing authority and time limits to better isolate the effect of legal oversight on attitudes.

Second, the study relies on self-reported perceptions and attitudes, which are inherently subject to social desirability bias, memory limitations, and individual interpretation of survey items. Even though the survey was anonymous, participants may have under- or over-reported their acceptance or concerns about surveillance to align with perceived social norms or personal ideals. Moreover, the concepts of "surveillance" and "privacy" may hold slightly different meanings for respondents depending on cultural background, political views, or prior experiences, introducing variability that is difficult to fully capture with standardized wording.

Third, while the sample was recruited to approximate demographic diversity, it does not perfectly represent the broader U.S. population. Certain groups (such as individuals with lower internet literacy, very low income, or limited English proficiency) may be underrepresented. This limits the generalizability of the findings and means that relationships observed here should be interpreted as indicative rather than definitive for the entire population.

Fourth, the cross-sectional design provides only a snapshot of attitudes at a single point in time. Surveillance practices, public discourse, and political climates can shift rapidly. Without longitudinal data, it is not possible to determine whether the relationships identified are stable over time or how attitudes might change in response to emerging technologies, new regulations, or major political events.

Fifth, although multiple statistical tests were conducted with appropriate corrections, the complexity of the dataset introduces a risk of both Type I and Type II errors. Small effect sizes may not have been detected, while some significant findings could still be spurious despite correction. Additionally, the reliance on non-parametric methods limits the ability to model more complex interactions or causal pathways.

Finally, practical constraints in survey implementation limited the ability to explore some promising avenues. For example, while demographic variables were extensive, factors such as prior personal experiences with surveillance or detailed political engagement were only indirectly measured. Such unobserved variables could moderate or mediate the relationships identified and remain an area for future research.

Taken together, these limitations highlight that the results should be viewed as an initial exploration rather than a definitive statement on public perceptions of digital and physical surveillance. Future work could build on these findings by incorporating proper tracking of experimental manipulations, expanding recruitment strategies to reach harder-to-survey populations, and employing longitudinal or mixed-methods designs to deepen understanding of the observed patterns.

7. CONCLUSION

This study set out to examine public perceptions of surveillance in the United States, comparing digital and physical methods in terms of perceived effectiveness, invasiveness, and acceptability. Additionally, it investigated the influence of institutional trust, demographic variables, and particularly political affiliation on these perceptions. The findings contribute to a growing body of research on the social legitimacy of surveillance practices in democratic societies and offer insights into the psychological and political factors that shape public attitudes.

A consistent pattern emerged across quantitative and qualitative responses: overt surveillance was more accepted than covert surveillance, and physical surveillance slightly more than digital. Yet, participants rated digital surveillance as more effective, aligning with beliefs that digital tools allow for broader data access, faster analysis, and greater predictive capacity. The tension between acceptability and effectiveness was most notable for digital covert methods, which were widely perceived as effective but also deeply invasive and ethically questionable.

Importantly, perceptions of invasiveness were less polarized than expected. A majority of participants rated digital and physical surveillance as equally invasive, and follow-up analysis revealed that question phrasing significantly influenced responses. Participants who were asked if digital surveillance was “more invasive” were more likely to agree than those asked the reverse. This framing effect suggests that perceptions of invasiveness are malleable and sensitive to subtle contextual cues. Still, participants consistently chose the surveillance method they perceived as less invasive, indicating that while invasiveness may not dominate evaluations, it still meaningfully guides preferences.

By contrast, perceived effectiveness was a far stronger predictor of surveillance acceptability. Participants were more willing to accept surveillance when they believed it would effectively prevent crime or promote safety. This finding was mirrored in open-text responses, where many acknowledged privacy trade-offs but justified certain methods as “necessary” or “justifiable” under the right conditions. This aligns with prior research suggesting that security concerns often override abstract privacy values when respondents are asked to consider specific, high-stakes scenarios.

A striking finding emerged when comparing institutional trust levels with those reported by Geierhaas et al. (2024). Across all four dimensions (trust in the government, government agencies, companies, and law enforcement) trust was significantly lower in the current sample. For instance, only 31.4% of respondents agreed that they trusted the government to do the right thing, compared to 41.8% in Geierhaas’ 2024 data. Similar drops were seen for trust in government agencies (down from 48.7% to 37.7%) and companies (from 50.8% to 43.8%). The most trusted institution remained law enforcement, yet even that figure dropped from 58.5% to 48.1%. These statistically significant changes may reflect a growing public disillusionment with institutions and heightened concern about data misuse, political overreach, or perceived erosion of civil liberties.

The survey also highlighted a deep political divide in attitudes toward surveillance. Political affiliation consistently emerged as one of the strongest predictors of trust, perceived effectiveness, and acceptance. Republicans and Democrats diverged not only in their levels of institutional trust but also in their perceived purpose and risks of surveillance. Participants with conservative views were generally more accepting of surveillance when positioned as a tool to fight crime or disorder, whereas liberal-leaning participants voiced stronger concerns about overreach, abuse of power, and the erosion of civil liberties. This aligns with broader societal trends where surveillance is viewed either as a tool of national protection or, conversely, as a symbol of authoritarian control, depending heavily on who holds power.

Notably, this political polarization extended beyond general attitudes to influence interpretation of specific methods. Technologies like social media monitoring, AI-based profiling, and client-side scanning were particularly controversial. Several respondents, particularly those expressing liberal or libertarian

views, questioned the legitimacy of these methods even when implemented overtly, citing algorithmic bias, chilling effects on speech, and lack of transparency. In contrast, conservative participants tended to focus more on whether surveillance served a functional purpose and were more likely to accept invasive methods if framed in the context of crime prevention or national security. This tension is particularly relevant in a politically fragmented society, where surveillance tools may be seen either as protective or oppressive depending on who is perceived to wield them.

At the same time, participants across the political spectrum expressed a common concern: surveillance is widespread and likely to be abused, but most do not believe they themselves will be affected. While the majority agreed that surveillance is now a routine part of life and increasingly hard to escape, only 15% believed they were personally likely to be surveilled. This gap between perceived prevalence and personal risk may help explain why surveillance remains widely tolerated despite widespread concern, it is seen as something that happens to “others.”

In conclusion, this study illustrates the complex and often contradictory nature of public attitudes toward surveillance. People value privacy but may sacrifice it for perceived security. They trust institutions less, yet still allow them considerable leeway, especially if surveillance is transparent or targeted. Most notably, attitudes are not formed in a vacuum but shaped by broader political narratives, individual experiences, and cultural expectations of what constitutes safety, fairness, and legitimacy.

As surveillance practices become increasingly embedded in both public and private life, these findings underscore the importance of transparency, accountability, and public dialogue. Without these, the legitimacy of surveillance risks being undermined, not only by technical overreach but by the erosion of trust across ideological lines. Future research should further explore the dynamics between political trust, social identity, and surveillance acceptance, particularly in multiparty democracies where shifting coalitions may complicate public expectations of oversight and power.

8. FUTURE WORK

Building on the findings of this study, several avenues for future research are evident. One important direction concerns the legal and contextual justification of surveillance. While this survey captured broad public attitudes, many participants voiced conditional acceptance of surveillance, hinging on whether a warrant was obtained or the method was used in serious cases (e.g. terrorism or child exploitation). Future studies should therefore investigate what the public considers a valid threshold for different types of surveillance. This could involve experimental vignettes that vary factors such as crime severity, location (public vs. private), or whether a judge approved the action, to assess how these conditions shift support.

Additionally, the distinction between types of surveillance and their intended targets warrants deeper exploration. Participants often emphasized that surveillance aimed at criminal suspects was more acceptable than blanket monitoring of the general population. Future research should disentangle perceptions of targeted vs. mass surveillance, and explore how people evaluate different combinations of method and context (e.g., facial recognition in crowds vs. on known suspects, or GPS tracking with vs. without consent).

Another key consideration is the impact of the political climate, particularly in the run-up to and aftermath of elections. The survey revealed a significant partisan divide in attitudes toward trust and surveillance acceptance. Participants often tied their responses to perceptions of the current government's legitimacy or fear of abuse under opposing leadership. As such, the 2024 U.S. presidential election and subsequent policy changes may significantly alter public attitudes. A follow-up study conducted before and after a major election cycle could illuminate how shifts in political power influence acceptance, fear, and trust in surveillance infrastructures.

A more robust understanding of oversight mechanisms is also needed. Several participants noted that they would support specific methods only if there was proper oversight, independent monitoring, or transparency about how data was used. Future work should investigate what kinds of oversight bodies the public actually trusts, and whether institutional reforms—like independent courts or citizen watchdog panels—could increase legitimacy in the eyes of skeptical groups.

Additionally, this study observed some confusion and framing sensitivity in comparative questions—particularly regarding invasiveness. Many participants answered based on the order or wording of the question, suggesting that people may lack a stable internal hierarchy between digital and physical surveillance. Future research could apply cognitive interviews or split-sample surveys to better capture how people conceptualize terms like "invasiveness" and to clarify how language and framing influence public opinion.

Lastly, future work could integrate policy literacy assessments. The study revealed that many participants hold complex, sometimes contradictory views on surveillance. Understanding whether this stems from lack of knowledge, distrust, or value-based reasoning would allow future research—and policymakers—to better design communication and regulation that aligns with both democratic norms and public expectations.

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9. APPENDIX

A. QUESTIONNAIRE

This is an overview of the full question set asked to the participants. In square brackets [] the answer options are listed. The round brackets () give more details, like if there is randomization. If there are bullet points with -, then these are suboptions, like in a question matrix. Conditional questions and when they are triggered is listed.

Welcome!

We invite you to take part in a research study by the University of Bonn on opinions about physical vs. digital surveillance. The survey takes about 10 minutes. No prior knowledge is needed.

Voluntary Participation

Your participation is completely voluntary. You can stop at any time by closing your browser—no explanation needed.

Your Answers

There are no right or wrong answers. We're only interested in your personal opinion. Please read the questions carefully and answer honestly.

Data Use and Privacy

We will collect your data anonymously and only with your consent. All data will be encrypted, stored securely, and used only for research and teaching. It may be included in scientific publications, but never in a way that identifies you. The survey is hosted on Qualtrics. We will ask for information like age, gender, nationality, profession, salary, and political preferences.

Benefits

You'll have a chance to share your views on surveillance and contribute to research that may inform future technologies or policies.

Who Is Behind This Study?

The study is conducted by Prof. Dr. Matthew Smith and Charlotte Mädler (B.Sc.), Institute of Computer Science, University of Bonn. Only our team will access the data.

Questions or Concerns? Contact us at charlotte.maedler@uni-bonn.de

You must consent to the collection and processing of data in order to participate in this study.

0. I consent to the collection and processing of my data as described. I confirm that I am at least 18 years old and that I want to participate in the study. [I consent, I do not consent]

There are two types of surveillance: covert (secret), where the target is unaware of being watched, and overt (non-secret), where the target is informed and can be present during searches.

Surveillance can target either digital spaces (smartphones, computers, online activity, etc.) or physical spaces (homes, possessions, public spaces, etc.).

Examples:

Overt Physical: Public police security cameras in plain sight

Covert Physical: Undercover police officers tailing a suspect

Overt digital: the government announces monitoring of social media posts related to an upcoming protest

Covert Digital: the police scan smartphone messages of an individual without their knowledge

1. Should US law enforcement agencies have the right to perform the following methods of surveillance in the USA? [Strongly Disagree, Somewhat Disagree, Neither, Somewhat Agree, Strongly Agree]

-Covert Digital

-Overt Digital

-Covert Physical

-Overt Physical

2. How effective do you think are the following surveillance methods at preventing crime or apprehending criminals? [Strongly Disagree, Somewhat Disagree, Neither, Somewhat Agree, Strongly Agree]

-Covert Digital

-Overt Digital

-Covert Physical

-Overt Physical

3(a).Surveillance of the digital space feels _____ compared to the physical space. [More invasive, equally invasive, less invasive]

3(b).Surveillance of the physical space feels _____ compared to the digital space. [More invasive, equally invasive, less invasive]

4.How often do you consciously think about your digital or physical privacy in everyday life? [Never, Rarely, Occasionally, Often, Constantly]

5. If you had to choose: would you prefer to be digitally or physically surveilled? [Digital, Physical, Both are equal to me](*randomized answer options*)

6. With which method do you think more information can be gathered about you, by US law enforcement agencies? [Digital, Physical, Both equally] (*randomized answer options*)

7. How likely do you think is it for physical surveillance to be abused by US law enforcement agencies?[Extremely unlikely, Somewhat unlikely, Neither likely nor unlikely, Somewhat likely, Extremely likely]

8.How likely do you think is it for digital surveillance to be abused by US law enforcement agencies ? [Extremely unlikely, Somewhat unlikely, Neither likely nor unlikely, Somewhat likely, Extremely likely]

9. On a scale of 0-10, how common in general do you think is surveillance in the USA by US law enforcement agencies ? [Slider 0-10](*randomized statement order*)

-Digital

-Physical

10. In the next 12 months, do you believe you could become a target of surveillance by US law enforcement agencies? [Very unlikely, Unlikely, Neither likely nor unlikely, Likely, Very likely]

11. Which surveillance type do you think would be more likely for your surveillance? [Digital, Physical, Both equally] (*randomized answer order*)

12. In the past 5 years, do you believe overall surveillance by US law enforcement agencies has increased, decreased or stayed the same? [Decreased significantly, decreased slightly, stayed the same, increased slightly, increased significantly]

(Condition: Q12: Increased slightly, increased significantly selected)

12.2 Do you believe digital or physical surveillance was increased more? [Digital, Physical, Both equally]

(*Randomized answer order*)

(Condition: Q12: Decreased slightly, decreased significantly selected)

12.3 Do you believe digital or physical surveillance was decreased more? [Digital, Physical, Both equally]

(*Randomized answer order*)

13. Do you find this worrisome? [Yes, A little, No]

14. Where you live, as how high do you perceive the following issues? [Very low, low, medium, high, very high] (*randomized statement order*)

-Crime Rate

-Corruption Rate

-Organized Criminality

15. In your opinion, does the government use surveillance methods more to protect you or limit personal freedoms in your area? [Protect, Limit, Neither, Both]

16.To what extent do you agree or disagree with the following statement: "In general, I trust the government to do the right thing." [Strongly agree, Agree, Neither agree nor disagree, Disagree, Strongly disagree]

17. To what extent do you agree or disagree with the following statement: "In general, I trust government agencies to comply with the law." [Strongly agree, Agree, Neither agree nor disagree, Disagree, Strongly disagree]

18. To what extent do you agree or disagree with the following statement: "In general, I trust companies such as Google and Apple to comply with the law." [Strongly agree, Agree, Neither agree nor disagree, Disagree, Strongly disagree]

AC. We ask questions to ensure participants are reading carefully. Please select 'Physical' below. [Digital, Physical, Both equally] (*Randomized Answer Order*)

19. To what extent do you agree or disagree with the following statement: "In general, I trust that law enforcement officials will treat me fairly." [Strongly agree, Agree, Neither agree nor disagree, Disagree, Strongly disagree]

20. For the following surveillance methods, please state whether you agree or not that US law enforcement agencies should be allowed to use them. [Strongly disagree, Somewhat disagree, Neither agree nor disagree, Somewhat agree, Strongly agree] (*Randomized Statement Order*)

-Built in Backdoors (Intentional security vulnerabilities placed in software or hardware that allow hidden access to a system, created by the developer or the government.)

-GPS Tracking (e.g. via Car or Smartphone, using location data from existing devices or vehicles)

-Internet and Network Monitoring (Tracking website visits, browser history and data traffic)

-Video Surveillance in Private Spaces using existing Hardware (Webcams, Ring, etc.)

-Installing Malware

-Social Media Monitoring (Analyzing posts, messages, and interactions on platforms)

-AI Analysis in the Internet (Using AI-driven tools to analyze online behavior patterns, facial recognition of online posts, etc.)

-Client Side Scanning (Automatic scanning of all sent and received messages or files e.g. on Instagram for potentially harmful or illegal content)

-Undercover Agents and Informants

-Wiretapping and Audio Surveillance

-Planting Video Surveillance in Private Spaces (Secretly installing hidden cameras in private spaces, e.g. in homes)

-Planting GPS Trackers

-CCTV and Security Cameras in Public Spaces

-Facial Recognition in Public Places (Using AI-driven tools to analyze public security cameras)

-Police Body-Cams

-Drones and Aerial Surveillance

Thank you for your responses! We will now follow up with a short individual question section and some questions about you.

(*Conditional: Q1: Strongly disagree or Somewhat disagree selected and for the majority of corresponding methods (Q19) somewhat agree or strongly agree was selected*)

21.1 You indicated that while you generally disagree that US law enforcement should be allowed to use (Digital Covert / Digital Overt / Physical Covert / Physical Overt) surveillance, you agreed to most or all of the corresponding methods. Can you think of any reason why?

(*Conditional: Q1: Strongly agree or Somewhat agree selected and for the majority of corresponding methods (Q19) somewhat disagree or strongly disagree was selected*)

21.2 You indicated that while you generally agree that US law enforcement should be allowed to use (Digital Covert / Digital Overt / Physical Covert / Physical Overt) surveillance, you disagreed to most or all of the corresponding methods. Can you think of any reason why?

(*Conditional: Q5 and Q6: for both digital or for both physical was selected*)

21.3. You indicated that while you believe with digital / physical surveillance more information can be gathered about you, you would still prefer digital / physical surveillance. Can you explain any reasons why?

(Conditional: Q14 Crime: Medium, High or Very High was selected)

21.4 Do you think law enforcement lacks the necessary surveillance capabilities to deal with everyday crime?
[Yes, No, Unsure]

(Conditional: Q14 Organized Crime: Medium, High or Very High was selected)

21.5 Do you think law enforcement lacks the necessary surveillance capabilities to deal with organized crime?
[Yes, No, Unsure]

(Conditional: 21.4 or 21.5 Yes was selected)

21.6 Which surveillance methods do you think law enforcement is currently lacking or should have greater access to in order to better address organized and/or everyday crime? [Access to end-to-end encrypted messages (e.g. Whatsapp), Real-time access to private financial transaction data, Continuous location tracking via mobile phones, Expanded use of facial recognition in public spaces, Bulk data access from social media platforms, Other (free text), None of the above] *(Multiple options selectable)*

(Conditional: Q14 Corruption: Medium, High or Very High was selected)

21.7 Would you be more supportive of surveillance if it were monitored by an independent third party? [Yes, No, Wouldn't make a difference, I am already fully supportive]

(Conditional: Q14 Either Crime or Organized Crime: Medium, High or Very High and Corruption: Medium, High or Very High was selected)

21.8 What worries you more in everyday life? [Crime/Organized Crime, Corruption, Both, Neither]

22. Do you feel you belong to a minority? [Yes, No, Rather not say]

(Conditional: Q22: Yes selected)

22.1. Do you have the feeling that a minority to which you belong to is being deliberately disadvantaged (violence, injustice, discrimination,...) by the following parties ? [Strongly disagree, Somewhat disagree, Neither agree nor disagree, Somewhat agree, Strongly agree] (randomized statement order)

-The US Government

-US law enforcement agencies

-The people of your country

-People in other countries

23. Do you think your answers to this survey would be significantly different if the Democratic Party were currently in control of the US government? [Definitely not, Probably not, Might or might not, Probably yes, Definitely yes]

(Conditional: Q23: Probably yes or definitely yes selected)

23.1 How would your answers differ? [Equally, More, Less] *(randomized answer order)*

-I would support surveillance:

-I would trust the government:

-I would trust law enforcement:

-I would fear surveillance abuse:

23.2 (Optional) How else would your answers differ? [free-text]

Demographics

23. How old are you? [Under 18, 18-24 years old, 25-34 years old, 35-44 years old, 45-54 years old, 55-64 years old, 65+ years old]

24. What is your gender? [Male, Female, Non-binary, I would like to describe myself (free-text), I'd rather not say]

25. In which state do you currently reside? [50 States, I do not currently reside in the United States]

26. What is your nationality? *(Multiple Options Selectable)* [US American, Other: (free-text), I'd rather not say] 42.
How would you describe yourself? *(Multiple Options Selectable)* [White/caucasian, Hispanic/Latino/Latina/Spanish, Black or African American, Asian or Southeast Asian, Middle Eastern or North

African, American Indian or Alaskan Native, Native Hawaiian or Pacific Islander, Other: (free-text), I would prefer not to say]

27. Do you currently or have you ever worked in one or more of the following sectors? (or are currently a student in one of these fields) (*Multiple Options Selectable*) [Government Employee, Military, Law Enforcement, IT/Tech, Cyber Security, None of the above, I prefer not to answer]

28. How high would you rate your tech-affinity? (0 being very low, 100 being very high) [Slider 0-100]

29. If elections were to be held next week, which party would you vote for? [Democratic Party, Republican Party, Independant, Other: (free-text), I can not vote, I would not vote, I prefer not to answer]

30. What is the highest level of education you have completed? [Some highschool or less, Highschool Diploma or GED, Some college but no degree, Associates or technical degree, Bachelor's degree, Graduate or professional degree, prefer not to say]

31. What was your total household income before taxes during the past 12 months? [Less than 25.000, 25.000-49.999, 50.000-74.999, 74.000-99.999, 100.000-149.999, 150.000 or more, Prefer not to say]

Thank you for your time! If you have any remarks or wish to provide feedback, please do so here. [Free-text]

B. OSF REGISTRATION

The OSF Registration can be found here: <https://osf.io/6uwjr>

[Add New](#)[My Registrations](#)[Help](#)[Donate](#)[Join](#)[Login](#)

Digital and Physical Surveillance in Public Perception

[Public registration](#) ▾ [Updates](#) ▾[Metadata](#)

Study Information



Hypotheses

We will test the following confirmatory hypotheses:

H1: Participants will perceive differences in how much information digital and physical surveillance can gather.

H2: Participants will rate the invasiveness of digital and physical surveillance differently.

H3: Acceptability differs between overt (informed) and covert (uninformed) surveillance.

H4: The perceived invasiveness of a surveillance type is associated with its acceptability.

H5: The perceived effectiveness of a surveillance method is associated with its acceptability.

H6-8: Trust in government, law enforcement, and companies is positively associated with acceptance of surveillance measures.

H9-11: Trust in government, law enforcement, and companies is negatively associated with concern about surveillance misuse.

H12: Perceived levels of local crime and organized crime are associated with surveillance acceptance.

Design Plan

Study type

Observational Study - Data is collected from study subjects that are not randomly assigned to a treatment. This includes surveys, "natural experiments," and regression discontinuity designs.

Blinding

No blinding is involved in this study.

Is there any additional blinding in this study?

No response

Study design

This is a cross-sectional, between-subjects survey study conducted using Qualtrics as the survey platform and Prolific for recruiting a representative sample of 1,000 U.S. adults. Participants are randomly

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The study includes bot detection and at least one attention check to ensure data quality. Sensitive questions include a "Prefer not to answer" option to protect participant privacy.

No files selected

Randomization

Randomization is implemented at multiple levels to reduce order effects and formulation bias:

Group Assignment (Between-Subjects):

Participants are randomly assigned using simple randomization to one of two groups that differ in the order of question blocks. One group is presented with descriptions of surveillance methods before evaluating their acceptability; the other group sees the acceptability questions first. These two options are presented with equal frequency.

Question Order Randomization (Within-Subjects):

For categorical response options, the answer choices are randomized unless there is a natural order (e.g., Likert scales from "strongly agree" to "strongly disagree").

Wording Direction (Formulation Bias Control):

Directionally sensitive comparative questions (e.g., "Digital surveillance is more/equally/less invasive than physical surveillance") are presented in either direction at random, using simple randomization at the question level.

All randomizations are implemented through Qualtrics' built-in randomization tools, which use simple (non-stratified) random assignment without replacement.

Sampling Plan

Existing Data

Registration prior to creation of data

Explanation of existing data

N/A

Data collection procedures

Data will be collected through an online survey hosted on Qualtrics, with participants recruited via Prolific. The target sample consists of 1,000 U.S.-based adults, selected to be demographically representative of the U.S. general population in terms of age, gender, and ethnicity, using Prolific's built-in representative sampling tools.

Inclusion criteria:

- Age 18 or older
- Consent given
- Currently residing in the United States
- Fluent in English
- Passed Prolific's pre-screening for representative eligibility

Exclusion criteria:

- Participants who fail the attention check
- Participants flagged as bots or providing incomplete responses
- Participants who complete the survey in less than 1/3 of the median completion time

The survey takes approximately 10 minutes to complete. Each participant is compensated at a fair hourly rate, in line with or exceeding Prolific's minimum pay guidelines.

Recruitment and data collection are expected to be completed within less than a week, depending on participant flow and quality screening. All data will be reviewed after collection to exclude invalid responses before analysis.

No files selected

Sample size

The study will analyze data from 1,000 individual participants, all adults residing in the United States. Each participant completes the survey once, providing a single data point per variable measured.

Sample size rationale

This sample size was determined using G*Power to provide sufficient power (0.80) to detect small to medium effect sizes (e.g. $r=0.1$, $r=0.3$) in the key relationships, such as between trust in institutions and acceptance of surveillance methods, at the standard .05 alpha error probability. Additionally, a margin was added for answers that will be excluded from analysis.

Stopping rule

No response

Variables

Manipulated variables

This is an observational study with no experimentally manipulated variables.

No files selected

Measured variables

Perceived Information Gathering Capabilities: Categorical variable with 3 levels: [Digital more/ Physical more/ Both equally].

Perceived Invasiveness: Collapsed into 3 levels: [Digital more invasive / Physical more invasive / Equal].

Acceptability Ratings: 4 items (Covert Digital, Overt Digital, Covert Physical, Overt Physical) on a 5-point Likert scale; used to compute means for digital, physical, overt, covert, and overall surveillance acceptance.

Effectiveness Ratings: 4 items, matching prior categories (acceptability), measuring perceived effectiveness on a 5-point Likert scale.

Trust Variables: Trust in government, law enforcement, and companies; each rated on a 5-point Likert scale.

Concern About: Average of physical and digital abuse likelihood, each on a 5-point Likert scale.

Perceived Crime and Organized Crime: Two separate ordinal 5-point items

No files selected

Indices

No response

No files selected

Analysis Plan

Statistical models

H1: We will use a chi-square goodness-of-fit test to test whether the distribution of responses on the Perceived Information Gathering Capabilities variable significantly deviates from uniform expectations.

H2: A chi-square goodness-of-fit test will be used to determine whether responses on the Perceived Invasiveness variable are evenly distributed across the three categories.

H3: We will compute the mean Acceptability Ratings for overt and covert surveillance (averaging the two items in each category) and compare them using a paired t-test.

H4: For each participant, we will calculate a difference score between their mean Acceptability Ratings for digital vs. physical surveillance. We will then use a one-way ANOVA (or Kruskal-Wallis test) to test whether this difference score varies across the three groups defined by the Perceived Invasiveness variable.

H5: For each of the four surveillance types, we will calculate a Spearman correlation between the corresponding Effectiveness Rating and Acceptability Rating. We will apply Holm-Bonferroni correction across these 4 correlations to control for multiple comparisons.

H6-8: We will compute Spearman correlations between each of the three Trust Variables (government, law enforcement, companies) and the mean of the four Acceptability Ratings. We will apply Holm-Bonferroni correction across these 3 correlations to control for multiple comparisons.

H9-11: We will compute Spearman correlations between each of the three Trust Variables and the Concern About variable (average perceived likelihood of abuse of digital and physical surveillance). We will apply Holm-Bonferroni correction across these 3 correlations to control for multiple comparisons.

H12: We will compute two Spearman correlations: One between Perceived Crime and the mean Acceptability Ratings, One between Perceived Organized Crime and the mean Acceptability Ratings. We will apply Holm-Bonferroni correction across these 2 correlations.

No files selected

Transformations

No response

Inference criteria

No response

Data exclusion

No response

Missing data

In the survey, answers to all questions are mandatory. The survey can not be completed unless the data is complete.

Exploratory analysis

In addition to our preregistered confirmatory hypotheses, we plan to possibly explore the following questions and relationships in the dataset. These analyses are hypothesis-guided but will be interpreted as exploratory, and any results we report will be described as such.

Potential topics to explore are:

EH1: Participants will perceive differences in how common digital and physical surveillance are.

EH2: Preferences for surveillance method (digital vs. physical) are associated with perceived invasiveness.

EH3: The perceived likelihood of being surveilled is associated with acceptance of surveillance practices.

EH4: Belief that surveillance has increased, decreased, or stayed the same is associated with the acceptance of surveillance.

EH5: Participants who both believe surveillance has increased and say they are worried about this will differ in their acceptance score compared to other combinations of these two variables.

EH6: Belief that surveillance has increased is associated with a different likelihood of saying the evolution of surveillance the past five years is worrisome than believing it has stayed the same or decreased.

EH7: Reporting that one feels deliberately disadvantaged by the government or police is associated with trust in EH8: Among those who perceive both crime and corruption as high, support for surveillance will differ depending on whether corruption or crime is perceived as more concerning.

EH9: Perceived levels of corruption are associated with trust in authorities.

EH10: Among participants who perceive high corruption and report low acceptance of surveillance, the option of independent monitoring is associated with differences in support for surveillance compared to those who reported high acceptance

EH11: Some participants will express general disagreement with a surveillance type (e.g., covert digital) while supporting most corresponding specific methods, and vice versa (general agreement but rejecting specific methods).

EH12: Some participants will express a preference for being surveilled through the method (digital or physical) that they also believe would reveal more information about them.

EH13: Age, gender, and education level are associated with differences in acceptance, trust, and perceived invasiveness of surveillance methods.

EH14: Participants with higher income or higher education will differ in their acceptance of digital surveillance methods.

EH15: Participants whose stated political affiliation aligns with the party currently in power (Republican) will differ in their trust in government, acceptance of surveillance, and concerns about surveillance abuse compared to participants whose affiliation does not align with the governing party.

EH16: Among participants who indicate their responses would change if the Democratic Party were currently in power, the type and direction of reported change (e.g., trust, support, fear of abuse) will differ based on their stated political affiliation.

Other

Other

No response

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C. JUPYTER NOTEBOOK AND DATA

The jupyter notebook for the stastical analysis done can be found on Github: <https://github.com/CharlyMae0811/MasterThesis>